

1. Why ac not dc?

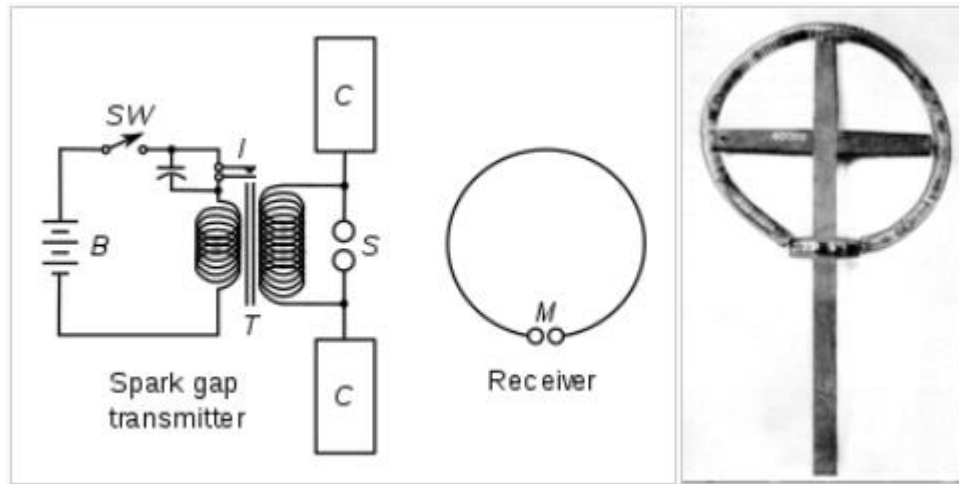
Watch *BBC: Shock and Awe: The Story of Electricity*



2. Regarding ac, we know the electron's drift speed is very low. If we reverse the direction of their motions very rapidly (60 Hz in US, 120 times/second) , the electrons can go nowhere. Then why electric devices still work?

Chap.33 Electromagnetic (EM) waves

Heinrich Rudolf Hertz (1857 –1894) was a German physicist who first conclusively **proved the existence of the electromagnetic waves** theorized by James Clerk Maxwell's electromagnetic theory of light. The unit of frequency — cycle per second — was named the "hertz" in his honor.



Hertz's 1887 apparatus



"It's of no use whatsoever... this is just an experiment that proves Maestro Maxwell was right—we just have these mysterious electromagnetic waves that we cannot see with the naked eye. But they are there."

Asked about the applications, Hertz replied, *"Nothing, I guess."*

Energy vs. wave number ~ thermal parameters

一杯茶，一包烟，
一道积分算一天。



Angular wave number : $k = 2\pi/\lambda$

Frequency: $f = \frac{1}{T} = \frac{\omega}{2\pi}$.

Angular frequency: $\omega = \frac{2\pi}{T}$.

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f.$$

$$0.03 \text{ THz} = 1 \text{ cm}^{-1}$$

$$1 \text{ meV} = 8.06 \text{ cm}^{-1}$$

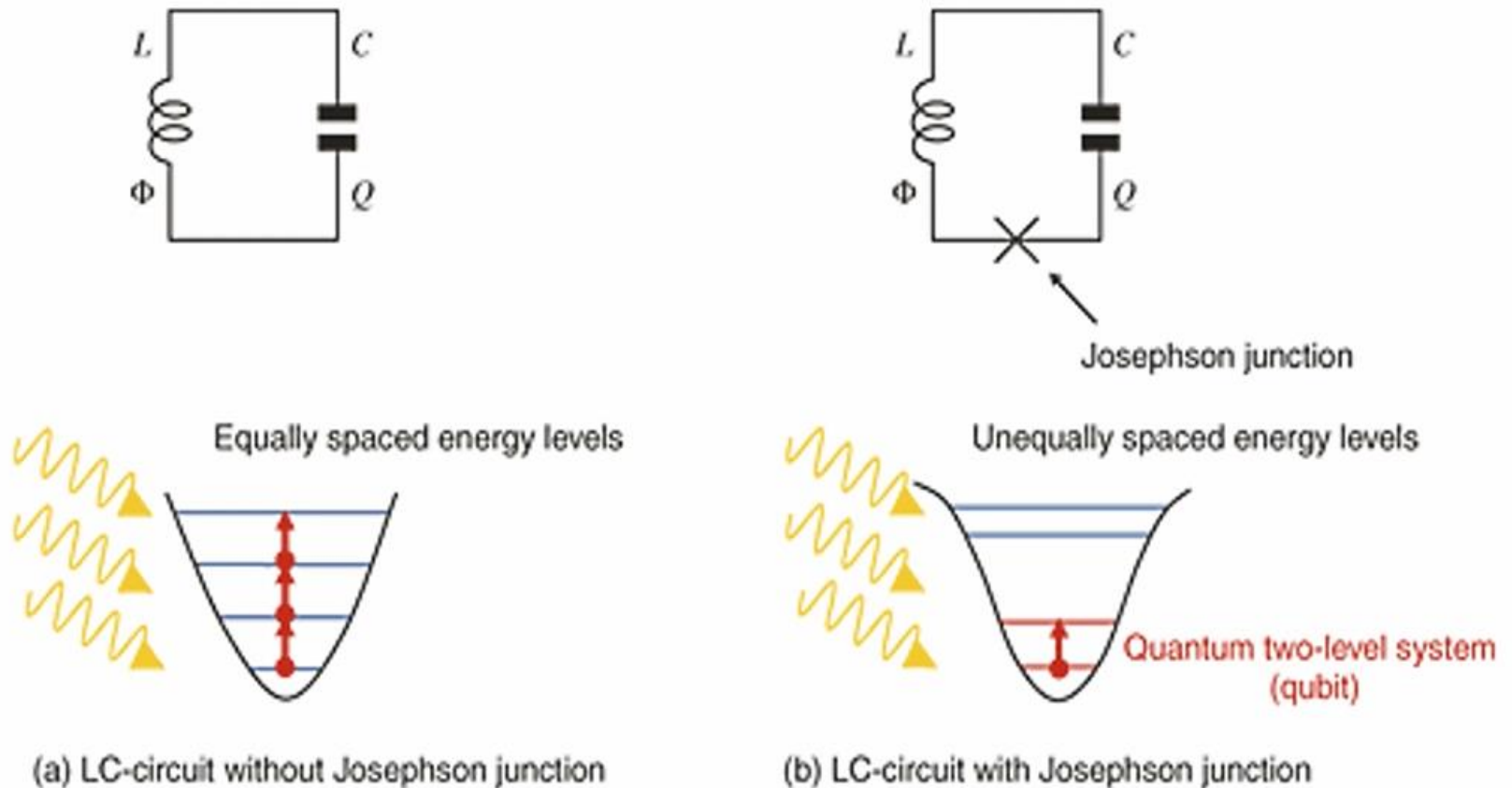
$$1.44 \text{ K} = 1 \text{ cm}^{-1}$$

$$1 \text{ meV} = 8 \text{ cm}^{-1} = 12 \text{ K} = 1 \text{ T} = 160 \text{ MPa}$$

<https://www.nature.com/immersive/d42859-019-00121-0/index.html>

Quantum LC circuit

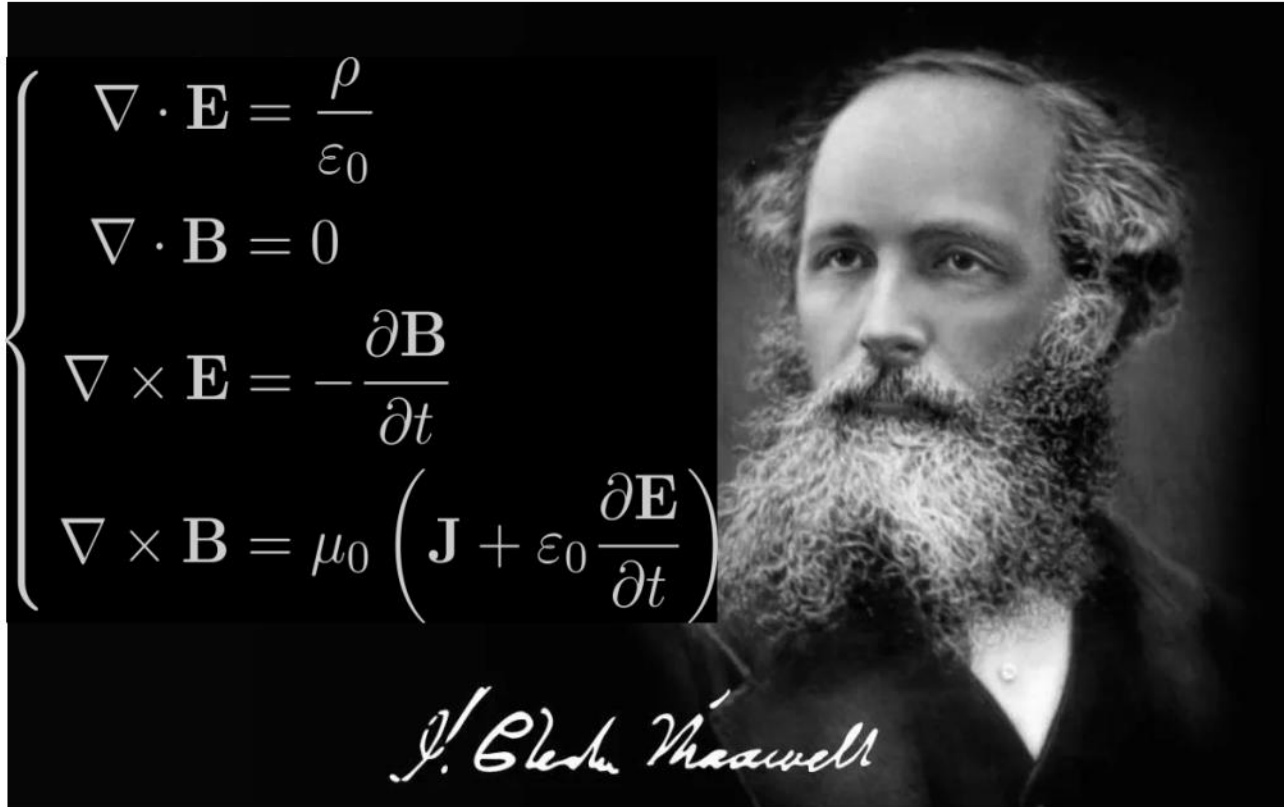
Superconducting quantum computing



Electromagnetic Waves

Chapter 33

Before Chapter 33, all we learned can be written as.....



静态情况:
电荷产生电场
(Gauss Law)
电流产生磁场
(Ampere Law)

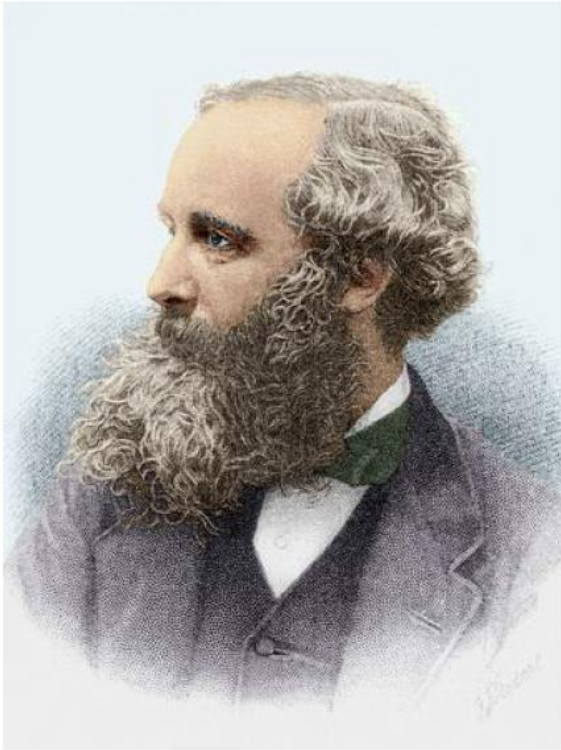
动态情况:
变化的电场可以产生磁场
(Maxwell-Ampere Law)
变化的磁场也可以产生电场
(Faraday Law)。

Light itself is an electromagnetic disturbance in the form of waves

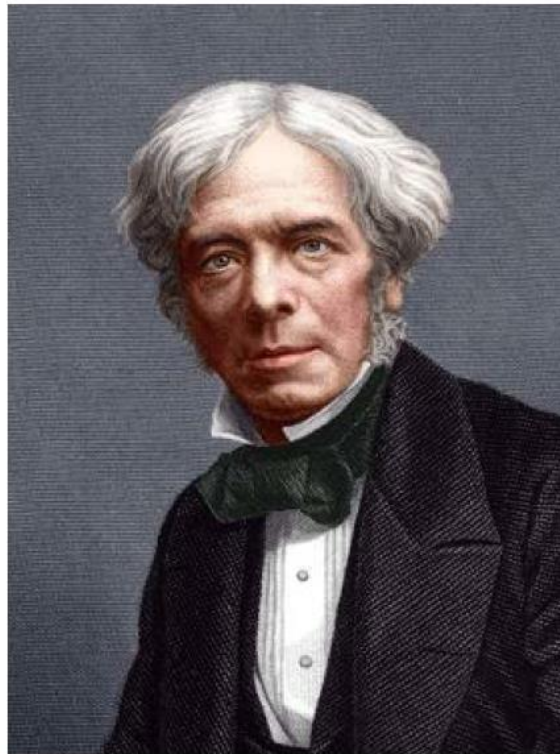
Electromagnetic Waves

- Electromagnetic Waves: Solution of Maxwell Equations
- **(光是一种电磁波)**
- Energy transportation & Radiation pressure (辐射压强)
- **(光能够携带能量和动量)**
- Polarization (极化) of lights
- **(光还有一种物理性质: 极化)**
- Reflection (反射) and refraction (折射)
- **(光在传播中会有反射和折射)**
- Chromatic dispersion (色散)
- **(折射会发生色散现象)**
- Polarization by reflection
- **(反射过程可能发生极化)**

When Maxwell meet Faraday



James C. Maxwell
(1831-1879)



Michael Faraday
(1791-1861)

当麦克斯韦征求他对论文的看法的时候，法拉第沉思地说，“但是，你不应该停留在用数学来解释我的观点，你应该突破它！”

A Unified Field Theory

- 1) unified all of the classical electromagnetism
- 2) directly indicates the electromagnetic nature of light

The magic of electromagnetic waves

Rainbow



X-ray



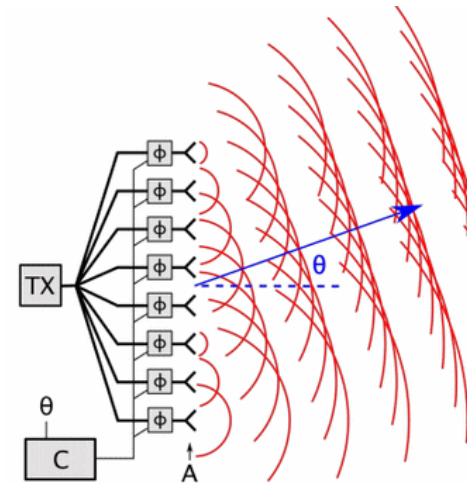
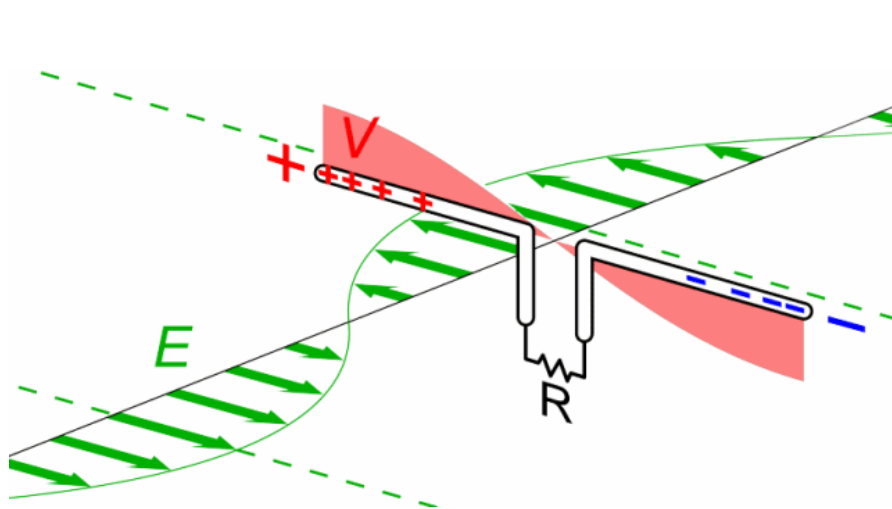
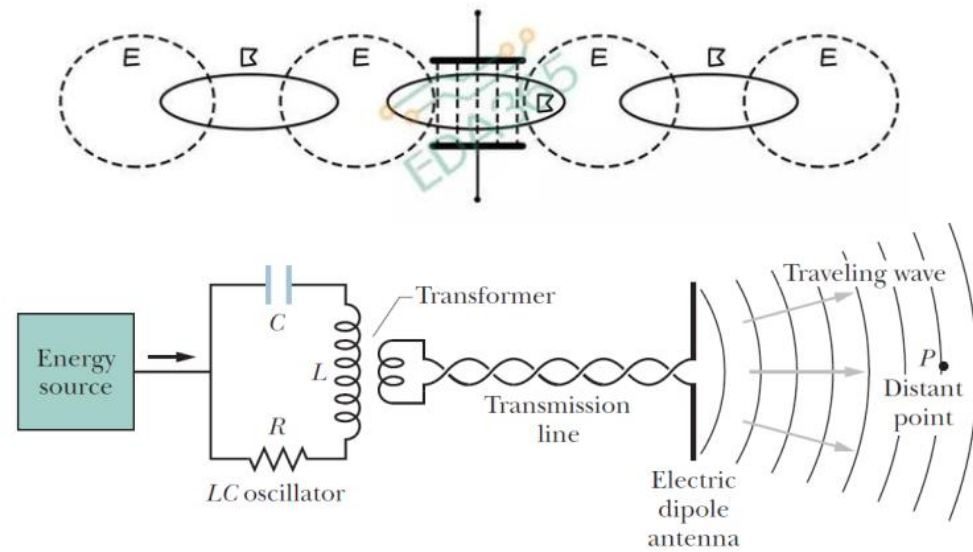
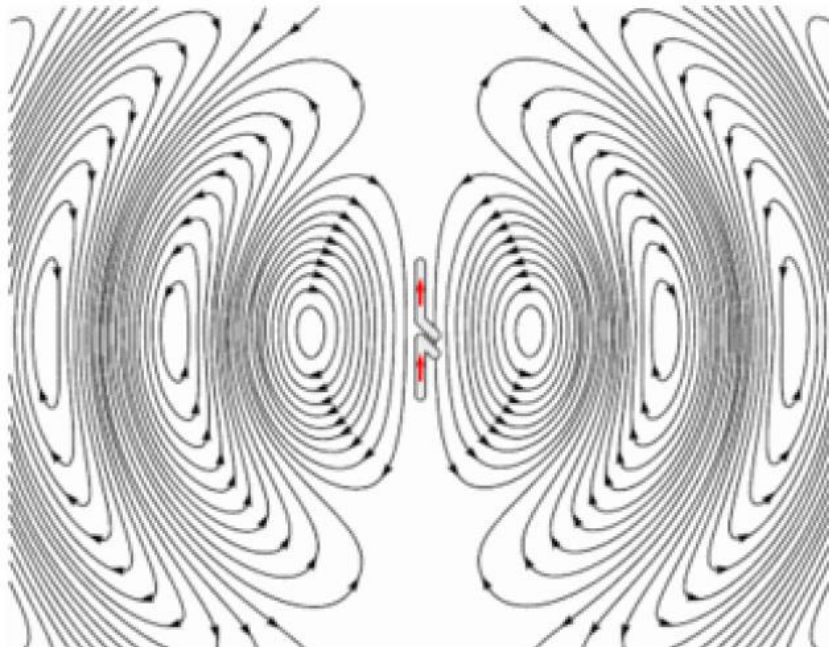
Laser



Radio



Appearance of EM wave



From Maxwell's equations to light*

Maxwell's equations

没有电荷, 没有电流,
只有相互诱导的电磁场

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right. \xrightarrow{\text{No charge, no current}} \left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = 0 \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{array} \right.$$

Take the curl of the fourth eq. and use the third eq.:

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

Use the known identity $\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B}$ on the left side, and since $\nabla \cdot \mathbf{B} = 0$, we get

$$\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0$$

$$\begin{aligned}
\nabla \times (\nabla \times \mathbf{A}) &= \nabla \times \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\
&= \nabla \times \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{e}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{e}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{e}_z \right] \\
&= \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) & \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) & \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \end{vmatrix} \\
&= \left(\frac{\partial^2 A_y}{\partial x \partial y} - \frac{\partial^2 A_x}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2} + \frac{\partial^2 A_z}{\partial x \partial z} \right) \hat{e}_x + \left(\frac{\partial^2 A_z}{\partial y \partial z} - \frac{\partial^2 A_y}{\partial z^2} - \frac{\partial^2 A_y}{\partial x^2} + \frac{\partial^2 A_x}{\partial x \partial y} \right) \hat{e}_y + \left(\frac{\partial^2 A_x}{\partial x \partial z} - \frac{\partial^2 A_z}{\partial x^2} - \frac{\partial^2 A_z}{\partial y^2} + \frac{\partial^2 A_y}{\partial y \partial z} \right) \hat{e}_z \\
&= \left(-\frac{\partial^2 A_x}{\partial x^2} - \frac{\partial^2 A_x}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2} \right) \hat{e}_x + \left(-\frac{\partial^2 A_y}{\partial y^2} - \frac{\partial^2 A_y}{\partial z^2} - \frac{\partial^2 A_y}{\partial x^2} \right) \hat{e}_y + \left(-\frac{\partial^2 A_z}{\partial z^2} - \frac{\partial^2 A_z}{\partial x^2} - \frac{\partial^2 A_z}{\partial y^2} \right) \hat{e}_z \\
&\quad + \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} \right) \hat{e}_x + \left(\frac{\partial^2 A_y}{\partial y^2} + \frac{\partial^2 A_z}{\partial y \partial z} + \frac{\partial^2 A_x}{\partial x \partial y} \right) \hat{e}_y + \left(\frac{\partial^2 A_z}{\partial z^2} + \frac{\partial^2 A_x}{\partial x \partial z} + \frac{\partial^2 A_y}{\partial y \partial z} \right) \hat{e}_z \\
&= -\nabla^2 A_x \hat{e}_x - \nabla^2 A_y \hat{e}_y - \nabla^2 A_z \hat{e}_z \\
&\quad + \hat{e}_x \frac{\partial}{\partial x} \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) + \hat{e}_y \frac{\partial}{\partial y} \left(\frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} + \frac{\partial A_x}{\partial x} \right) + \hat{e}_z \frac{\partial}{\partial z} \left(\frac{\partial A_z}{\partial z} + \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} \right) \\
&= -\nabla^2 \mathbf{A} + \nabla (\nabla \cdot \mathbf{A})
\end{aligned}$$

$$\begin{aligned}
&[\nabla \times (\nabla \times \mathbf{E})]_i \\
&= \epsilon_{ijk} \nabla_j (\nabla \times \mathbf{E})_k \\
&= \epsilon_{ijk} \nabla_j \epsilon_{kmn} \nabla_m E_n \\
&= \epsilon_{ijk} \epsilon_{kmn} \nabla_j \nabla_m E_n \\
&= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \nabla_j \nabla_m E_n \\
&= \nabla_n \nabla_i E_n - \nabla_m \nabla_m E_i \\
&= \nabla_i (\nabla \cdot \mathbf{E}) - (\nabla \cdot \nabla) E_i \\
&= [\nabla (\nabla \cdot \mathbf{E})]_i - (\nabla^2 \mathbf{E})_i
\end{aligned}$$

From Maxwell's equations to light*

Since the Maxwell's eqs. are symmetric on $\mathbf{E}$ and $\mathbf{B}$ when there's neither charge nor current (no media needed for EM waves to travel!!), we have

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}.$$
$$\begin{cases} \nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0 \\ \nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \end{cases} \quad \frac{d^2 x(t)}{dt^2} + \omega^2 x(t) = 0.$$

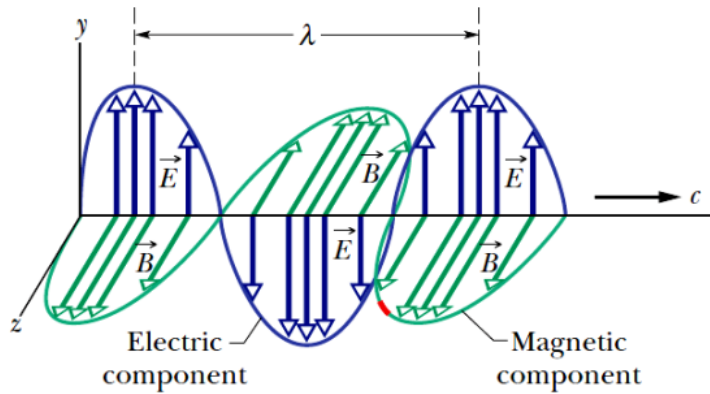
This is the complete form of the **electromagnetic wave equations** who describes all forms of EM waves (light) travelling at speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 299\,792\,458 \text{ m/s.}$$

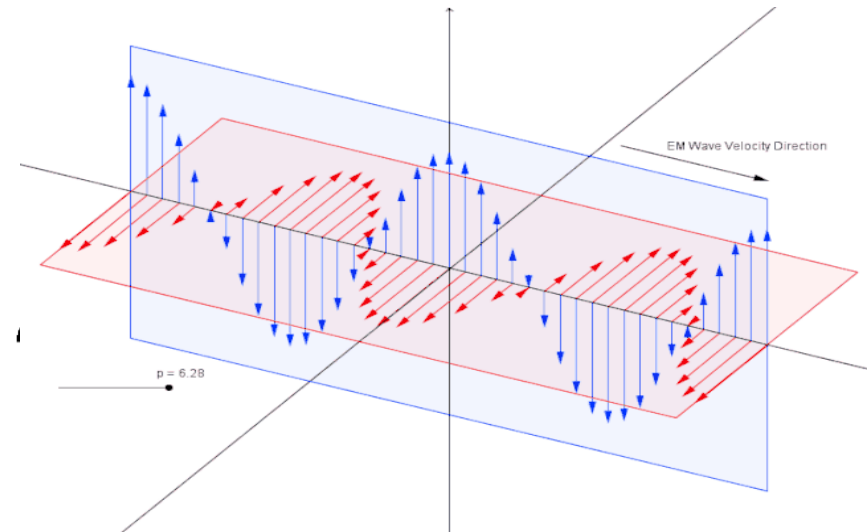
This velocity is so nearly that of light, we have strong reasons to conclude that light itself is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws

James Clerk Maxwell, 1865

Characterization of waves



空间周期性 (波长 λ)



$$\Delta\theta = 2\pi \frac{\Delta x}{\lambda} = \frac{2\pi}{\lambda} \Delta x = k\Delta x$$

$$k = \frac{2\pi}{\lambda}$$

相位周期性 (2π)

$$\Delta\theta = 2\pi \frac{\Delta t}{T} = \frac{2\pi}{T} \Delta t = \omega\Delta t$$

$$\omega = \frac{2\pi}{T}$$

时间周期性 (周期 T)

$$\frac{\lambda}{T} = c = \frac{\omega}{k}$$

$$\begin{cases} E = E_m \sin(kx - \omega t) \\ B = B_m \sin(kx - \omega t) \end{cases}$$

Properties of the EM wave

The general solutions are:

$$\begin{cases} \mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \phi_0) \\ \mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \phi_0) \end{cases} \xrightarrow{\text{In the book}} \begin{cases} E = E_m \sin(kx - \omega t) \\ B = B_m \sin(kx - \omega t) \end{cases}$$

where

t is time (in seconds),

ω is the angular frequency of the EM wave,

$\mathbf{k} = (k_x, k_y, k_z)$ is the **wave vector**, and

ϕ_0 is the phase angle.

The wave vector $\mathbf{k}$ is related to ω by

$$k = |\mathbf{k}| = \frac{\omega}{c} = \frac{2\pi}{\lambda}.$$

We can also derive that

$$\left| \frac{\mathbf{E}_0}{\mathbf{B}_0} \right| = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

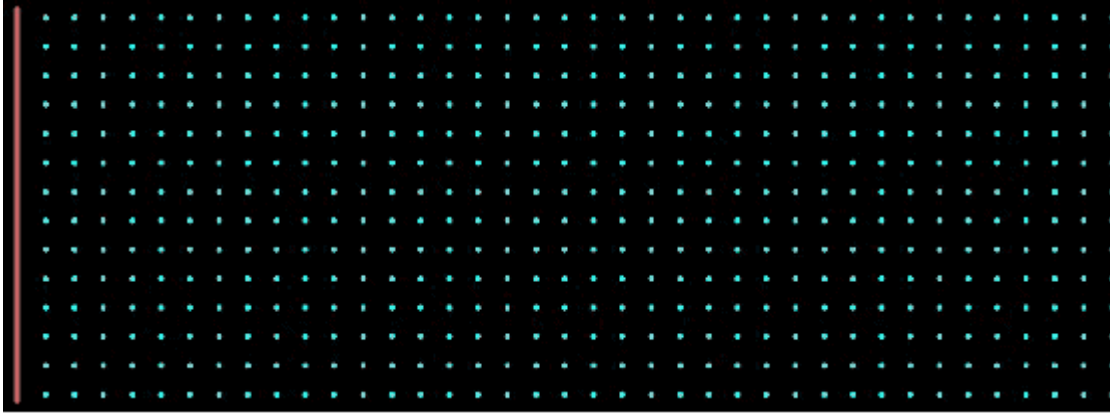
$$kx - \omega t = \phi$$

$$k\Delta x - \omega\Delta t = 0$$

$$c = \frac{\Delta x}{\Delta t} = \frac{\omega}{k}$$

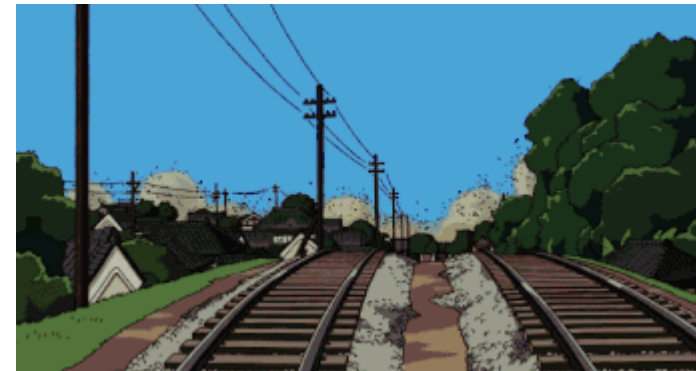
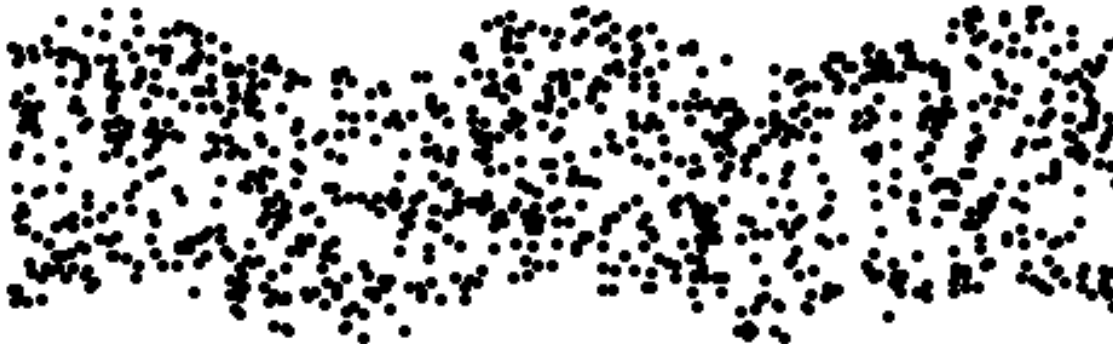
EM waves → Transverse waves

Longitudinal waves - transfer energy *in the same direction* as the disturbance in the medium of propagation

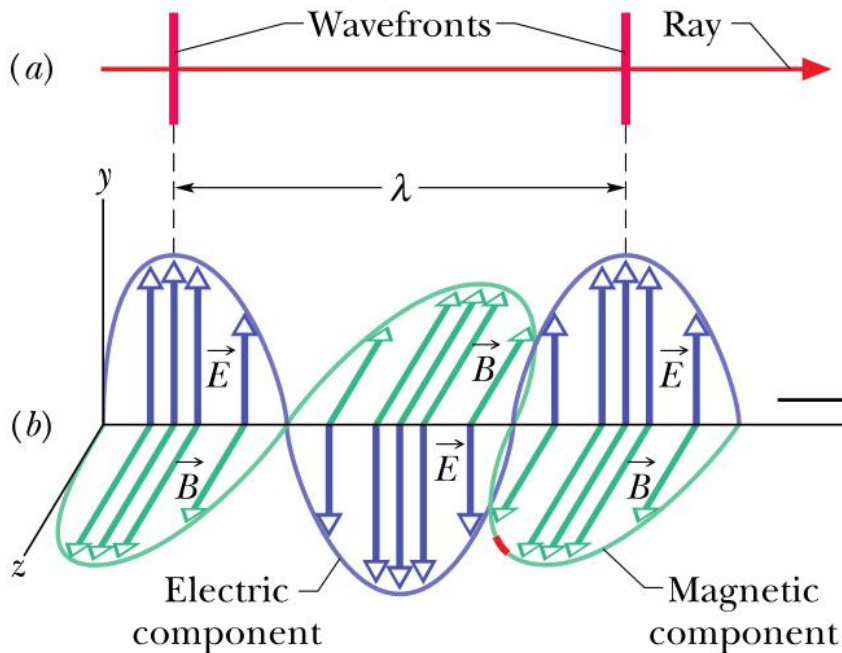


$$\frac{\partial^2 s}{\partial t^2} = \frac{B}{\rho} \frac{\partial^2 s}{\partial x^2},$$

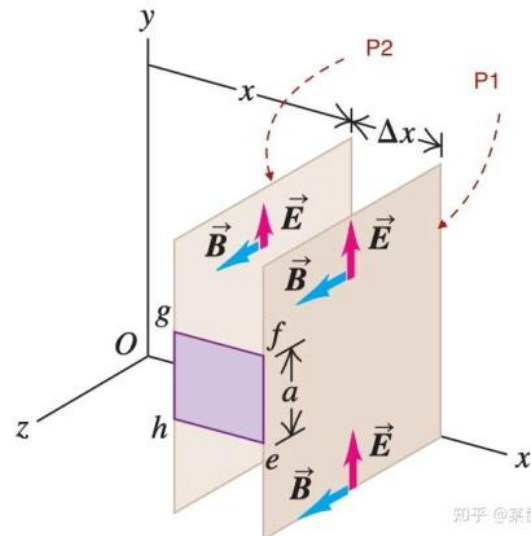
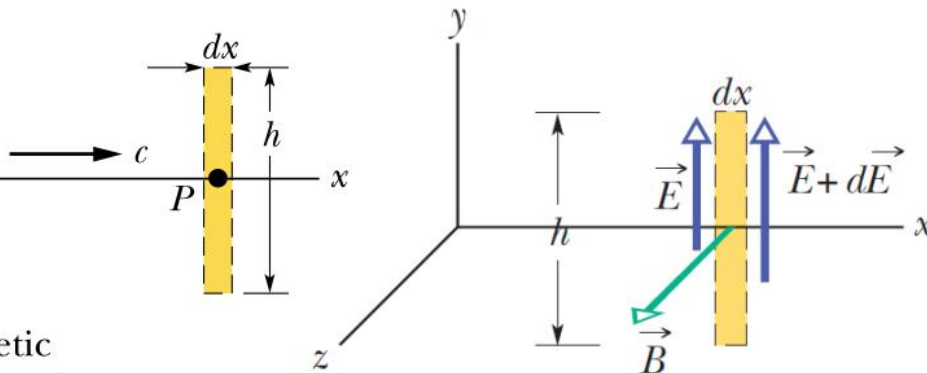
Transverse waves - transfer energy in a direction *perpendicular* to the direction of the disturbance in the medium



Oscillating B induces an oscillating E



The oscillating magnetic field induces an oscillating and perpendicular electric field.



Faraday's law $\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$

$$\oint \vec{E} \cdot d\vec{s} = (E + dE)h - Eh = h dE$$

$$\Phi_B = (B)(h \, dx) \qquad \frac{d\Phi_B}{dt} = h \, dx \frac{dB}{dt}$$

$$h \, dE = -h \, dx \frac{dB}{dt} \quad \frac{dE}{dx} = - \frac{dB}{dt}$$

Oscillating E induces an oscillating B

$$\begin{cases} E = E_m \sin(kx - \omega t) & \frac{\partial E}{\partial x} = kE_m \cos(kx - \omega t) \\ B = B_m \sin(kx - \omega t) & \frac{\partial B}{\partial t} = -\omega B_m \cos(kx - \omega t) \end{cases}$$

$$kE_m \cos(kx - \omega t) = \omega B_m \cos(kx - \omega t)$$

$$\frac{E_m}{B_m} = \frac{\omega}{k} = c \quad (\text{Amplitude ratio})$$

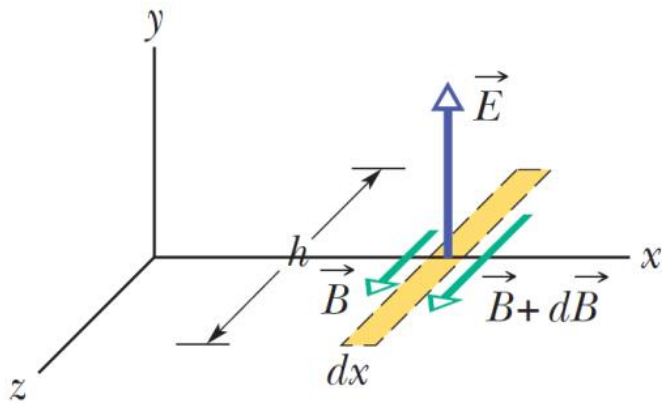
The oscillating electric field induces an oscillating and perpendicular magnetic field.

Ampere
-Maxwell law $\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$

$$-\frac{\partial B}{\partial x} = \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

$$-kB_m \cos(kx - \omega t) = -\mu_0 \epsilon_0 \omega E_m \cos(kx - \omega t)$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (\text{wave speed})$$



EM waves in matter

Remember
in Chapter 25 :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

(in vacuum)



$$E = \frac{1}{4\pi\kappa\epsilon_0} \frac{q}{r^2}$$

(in dielectric)

dielectric
constant

$$\epsilon_0 \Rightarrow K\epsilon_0 = \epsilon$$

Permittivity of
the dielectric

(in vacuum)

(in dielectric)

$$\mu_0 \Rightarrow K_m\mu_0 = \mu$$

Permeability of
the magnet

Relative permeability of
the magnet

Therefore the electromagnetic wave speed in dielectric (v) is :

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{1}{\sqrt{KK_m}} \frac{1}{\sqrt{\epsilon_0\mu_0}} = \frac{c}{\sqrt{KK_m}} \longrightarrow \frac{c}{\sqrt{K}}$$

For most dielectrics the relative
permeability $K_m \cong 1$

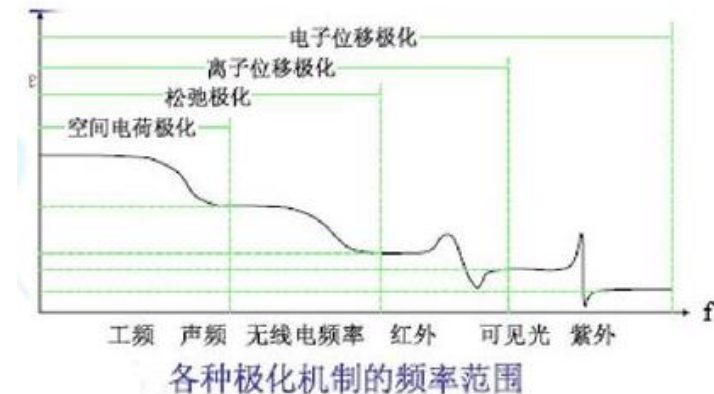
In Chapter 25 :

Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310

However, at rapid oscillating field at the **frequency of visible light**, K for water is 1.8

K is actually **a function of frequency** (the dielectric function)



Example

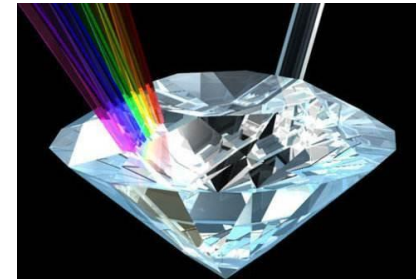
Visiting a jewelry store one evening, you hold a diamond up to the light of a sodium-vapor street lamp. The heated sodium vapor emits yellow light with a frequency of 5.09×10^{14} Hz.

Find the **wavelength in vacuum** and the **wave speed and wavelength in diamond**, for which $K = 5.84$ and $K_m = 1.00$ at this frequency.

$$c = \lambda f$$

The wavelength in vacuum of the sodium light is

$$\lambda_{\text{vacuum}} = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{5.09 \times 10^{14} \text{ Hz}} = 5.89 \times 10^{-7} \text{ m} = 589 \text{ nm}$$



The wave speed and wavelength in diamond are

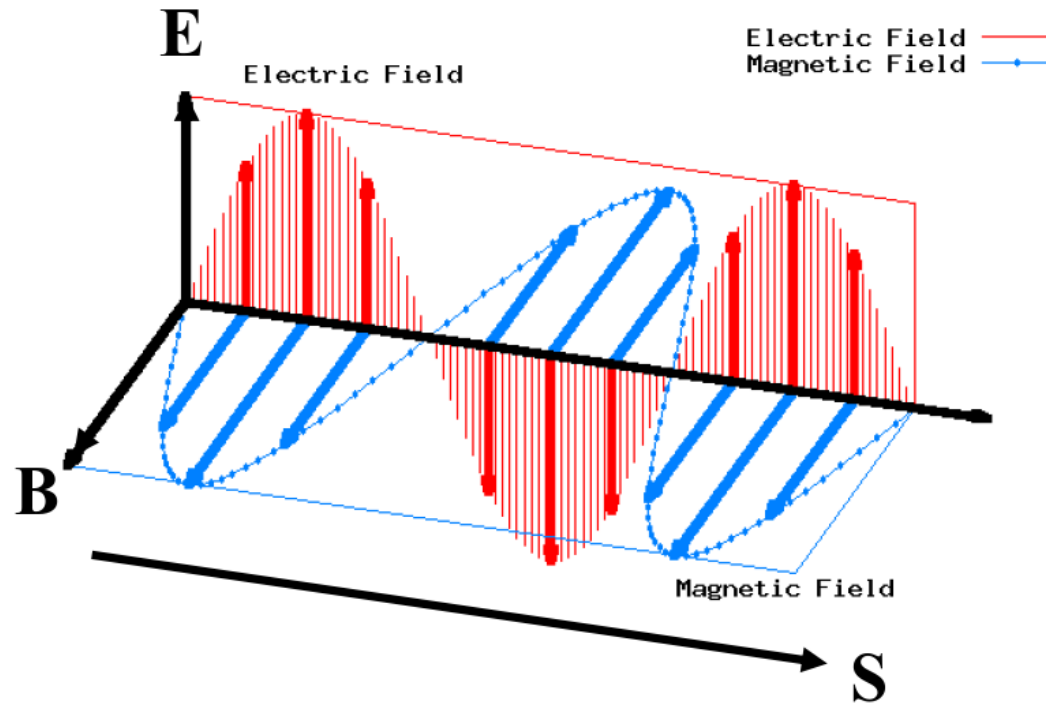
$$v_{\text{diamond}} = \frac{c}{\sqrt{KK_m}} = \frac{3.00 \times 10^8 \text{ m/s}}{\sqrt{(5.84)(1.00)}} = 1.24 \times 10^8 \text{ m/s}$$

$$\lambda_{\text{diamond}} = \frac{v_{\text{diamond}}}{f} = \frac{1.24 \times 10^8 \text{ m/s}}{5.09 \times 10^{14} \text{ Hz}} = 244 \text{ nm}$$

II. Energy transfer in EM waves

EM waves can transfer energy. The rate of energy transport per unit area is described by the **Poynting vector**

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}.$$



Energy transport and the Poynting vector

In an EM wave that travels through vacuum, $\mathbf{E}$ is perpendicular to $\mathbf{B}$, so

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} = \frac{1}{\mu_0} EB = \frac{1}{c\mu_0} E^2.$$

This is the instantaneous energy flow rate of the EM wave. The time average S is also called the **intensity I of the wave**:

$$I = \langle S \rangle = \frac{1}{c\mu_0} \langle E^2 \rangle = \frac{1}{2c\mu_0} E_m^2 = \frac{1}{c\mu_0} E_{\text{rms}}^2.$$

where $E_{\text{rms}} = \frac{E_m}{\sqrt{2}}$ root-mean-square 均方根

$$\langle E^2 \rangle = \langle E_m^2 \sin^2(kx - \omega t) \rangle = \frac{E_m^2}{2}$$

Energy transport and the Poynting vector

$$I = \langle S \rangle = \frac{1}{c\mu_0} \langle E^2 \rangle = \frac{1}{2c\mu_0} E_m^2 = \frac{1}{c\mu_0} E_{\text{rms}}^2.$$

In an EM wave, we also have (instantaneous values)

$$\frac{1}{2}\epsilon_0 E^2 = u_E = u_B = \frac{B^2}{2\mu_0}$$

$$\frac{E}{B} = c$$

能量密度

$$I = \left(\frac{1}{2} \epsilon_0 E_m^2 \right) c = \left(\frac{1}{2\mu_0} B_m^2 \right) c$$

波的强度：能量流密度

$$j = (ne)v_d$$

电（荷）流密度

Intensity and Power

	About Charge	About Energy
	Current (charge/time) $I = j A$	Power (energy/time) $P = I A$
	Current density (charge/time/area) $j = (ne)v_d$ Vector	Intensity [of the wave] (energy/time/area) $I = \left(\frac{1}{2} \epsilon_0 E_m^2 \right) c = \left(\frac{1}{2\mu_0} B_m^2 \right) c$ Vector (Poynting)
	Charge density (charge/volume) $\rho = ne$	Energy density (energy/volume) $u_E = \frac{1}{2} \epsilon_0 E^2, u_B = \frac{1}{2\mu_0} B^2$

Example : Solar Energy

The average intensity of noontime sunlight on a clear day is about 1 kW/m^2 .

- (a) What are the peak electric & magnetic fields in sunlight?
- (b) At this intensity, how large a 40% efficient solar collector would you need to replace a 4.8-kW water heater?



$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad S_{avg} = \frac{E_0 B_0}{2 \mu_0} \quad E_0 = c B_0$$

$$\begin{aligned} \text{(a)} \quad S_{avg} &= \frac{1}{\mu_0} c \frac{B_0^2}{2} & B_0 &= \sqrt{\frac{2 \mu_0 S_{avg}}{c}} = \sqrt{\frac{2 (4\pi \times 10^{-7} \text{ H/m}) (10^3 \text{ W/m}^2)}{3 \times 10^8 \text{ m/s}}} \\ & & &= 2.9 \times 10^{-6} \text{ T} = 2.9 \mu\text{T} \end{aligned}$$

$$E_0 = c B_0 = 8.7 \times 10^2 \text{ V/m} = 0.87 \text{ kV/m}$$

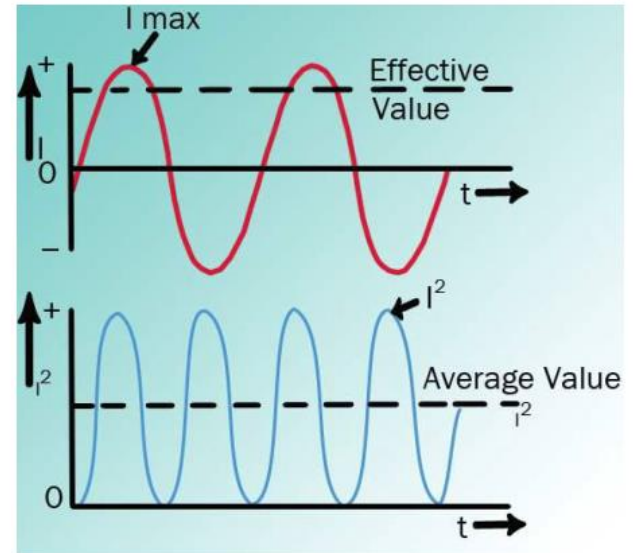
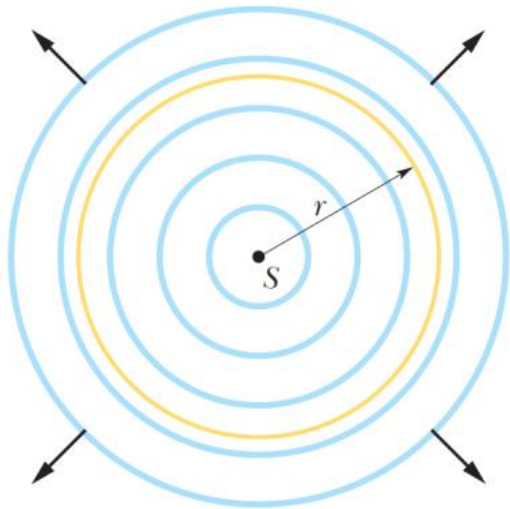
$$\text{(b)} \quad \text{Area needed is } \frac{4.8 \text{ kW}}{40\% \times 1.0 \text{ kW/m}^2} = 12 \text{ m}^2$$

Intensity and Power

$$S = \left(\frac{\text{energy/time}}{\text{area}} \right)_{\text{inst}} = \left(\frac{\text{power}}{\text{area}} \right)_{\text{inst}}$$

$$I = S_{\text{avg}} = \left(\frac{\text{energy/time}}{\text{area}} \right)_{\text{avg}} = \left(\frac{\text{power}}{\text{area}} \right)_{\text{avg}}$$

The energy emitted by light source S must pass through the sphere of radius r .



$$I = \frac{\text{power}}{\text{area}} = \frac{P_s}{4\pi r^2}$$

Example : Mobile Phone Reception

A mobile phone's typical average power output is about 0.6 W.

If the receiver at a cell tower can handle signals with peak electric fields as weak as 1.2 mV/m, what is the maximum allowable distance from mobile phone to tower ?

Change of **intensity** I with distance for a point source :

$$I = \frac{P_s}{4\pi r^2} = S_{avg} = \frac{E_0 B_0}{2 \mu_0} = \frac{E_0^2}{2 \mu_0 c}$$

$$S_{avg} = \frac{P_{avg}}{4\pi r^2} = \frac{1}{\mu_0} \frac{E_0^2}{2 c}$$

$$r = \sqrt{\frac{2 \mu_0 c P_{avg}}{4\pi E_0^2}} = \sqrt{\frac{2 (4\pi \times 10^{-7} \text{ H / m}) (3 \times 10^8 \text{ m / s}) (0.6 \text{ W})}{4\pi (1.2 \times 10^{-3} \text{ V / m})^2}} = 5 \text{ km}$$



Radiation pressure

We can also understand light as particles, which we called **photons**.

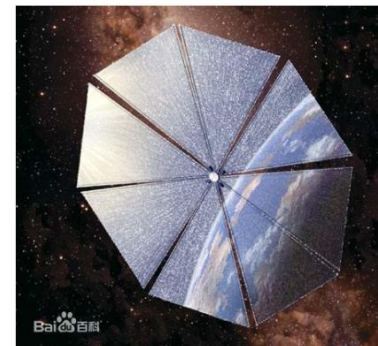
Photons have zero mass at rest, but they can carry **energy** and they have **momentum**

$$\text{Momentum } \mathbf{p} = \frac{\text{energy of the photon}}{c}$$

Therefore, light or EM waves can exert force / pressure on an object.

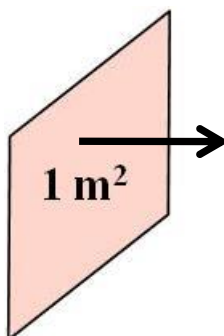
Force : rate of change of momentum

Pressure : Force / area



$$\text{Change of momentum } \Delta p = \frac{\text{change of energy}}{c} \quad (\text{totally absorbed})$$

$$\Delta p = 2 \frac{\text{change of energy}}{c} \quad (\text{totally reflected})$$

A diagram showing a light beam represented by an arrow pointing from left to right. The arrow is labeled with 'I' above and 'c' below. It is hitting a pink rectangular surface labeled '1 m²'.
$$\frac{I}{c} = \frac{S_{avg}}{c} = \frac{\text{energy}}{c \text{ m}^2 \text{ sec}} = \text{pressure} \quad (\text{with total absorption})$$
$$= \frac{2I}{c} \quad (\text{with total reflection})$$

We experience pressure when light shines on us!

Radiation pressure of a satellite

An earth-orbiting satellite has solar energy-collecting panels with a total area of 4.0 m^2 . If the sun's radiation (1.4 kW/m^2) is perpendicular to the panels and is completely absorbed, find the average radiation-pressure force.



Radiation Pressure

$$p_r = \frac{F}{A} = \frac{I}{c} \quad (\text{with total absorption})$$

$$= \frac{1400}{3 \times 10^8} = 4.67 \times 10^{-6} \text{ N/m}^2$$

$$\text{Force} = 4.67 \times 10^{-6} (4) = 1.87 \times 10^{-5} \text{ N}$$

as small as the weight of a single grain of salt, but still has to be taken into account for orbit calculation!

Radiation pressure

If the photon energy $E = pc$ is **totally reflected** (not absorbed) by an object, the object gains a linear momentum

$$\Delta p = \frac{2\Delta E}{c} = F\Delta t,$$

Therefore,

$$\Delta p = \frac{2\Delta E}{c} = \frac{2IA\Delta t}{c} = F\Delta t,$$

$$F = \frac{2IA}{c},$$

and the radiation pressure is now

$$p_r = \frac{F}{A} = \frac{2I}{c}.$$



光压风车

Radiation pressure on comet

Ion tail / plasma tail due to solar wind

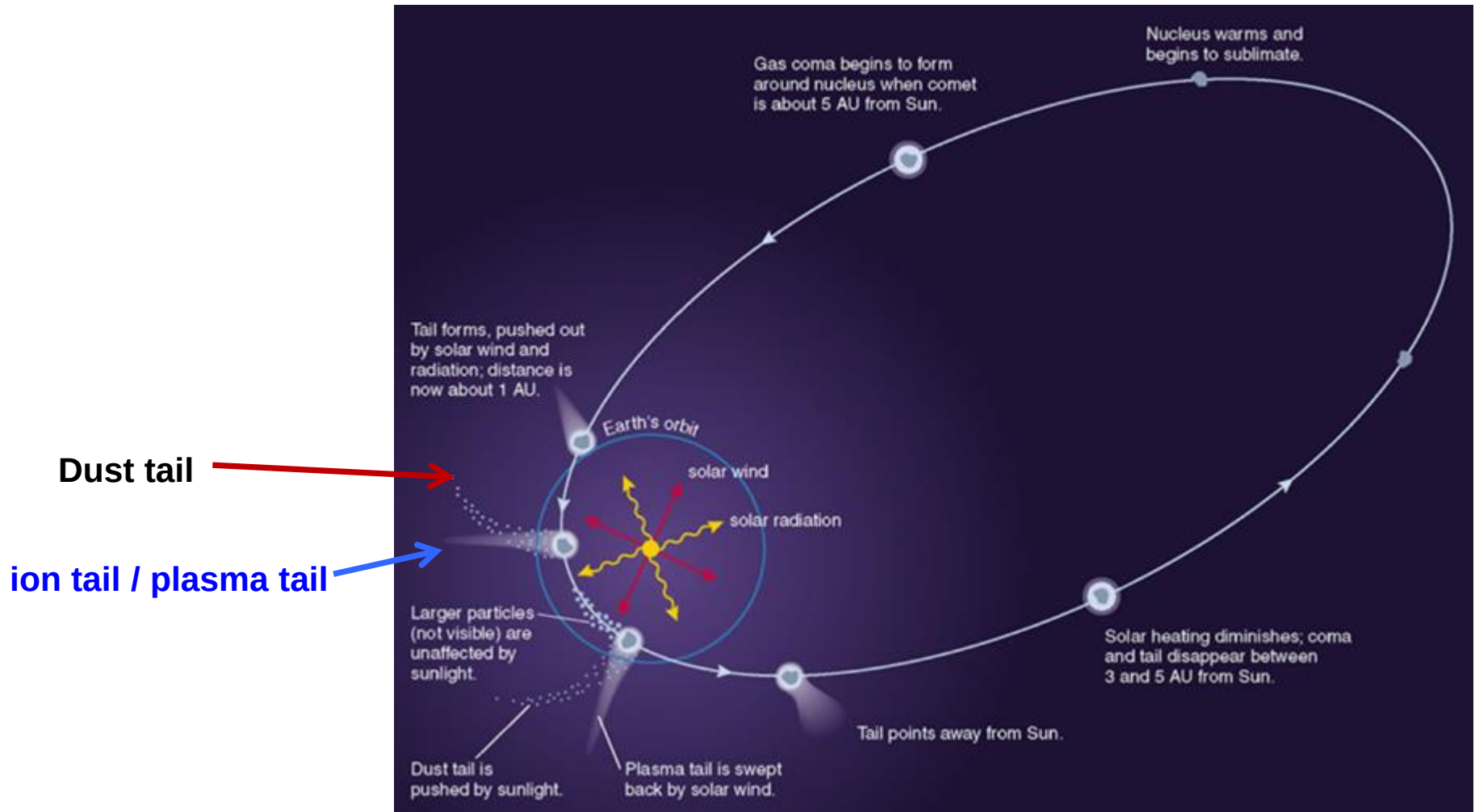
Direction : away from the sun

Dust tail due to radiation pressure

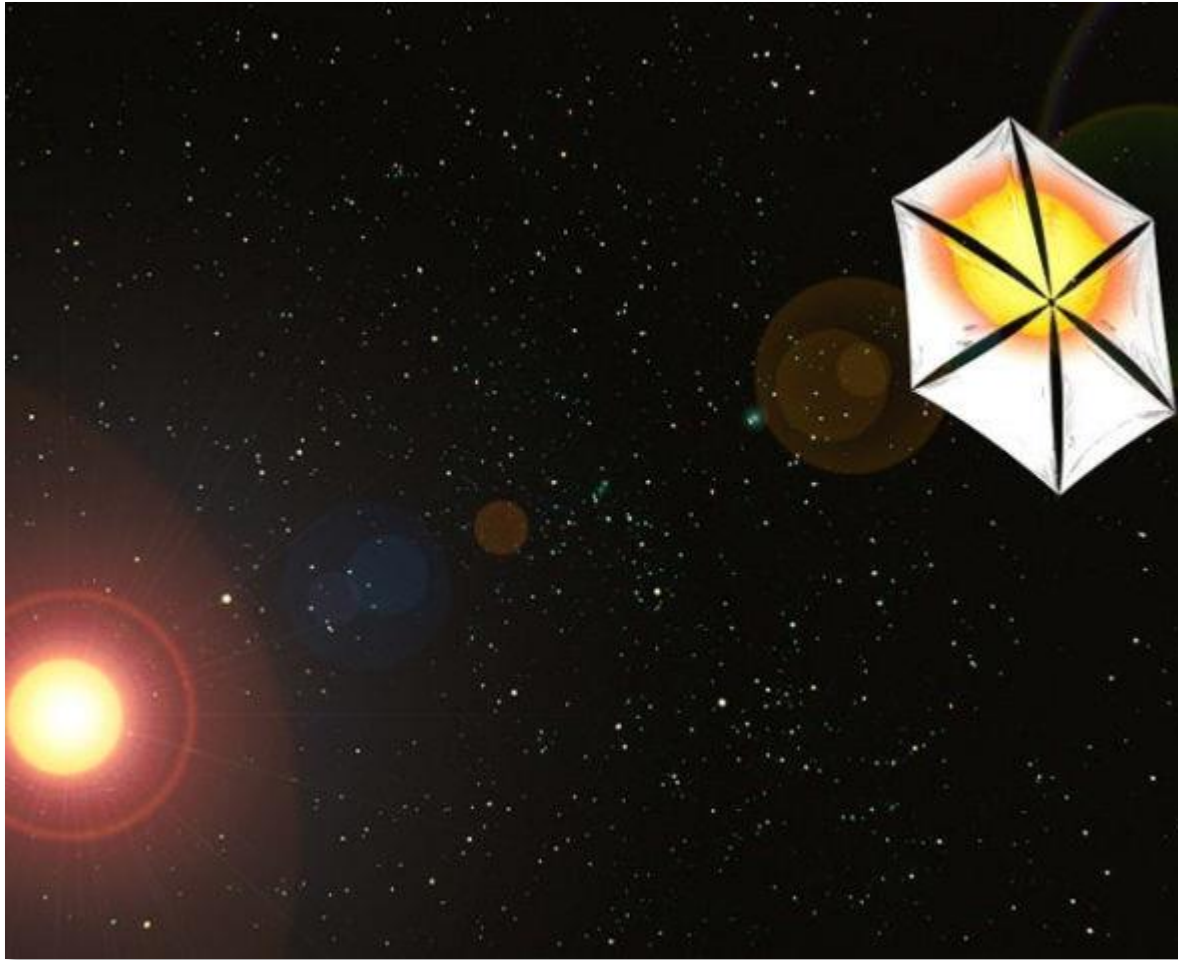
Direction : away from the sun and the orbit direction



Radiation pressure on comet



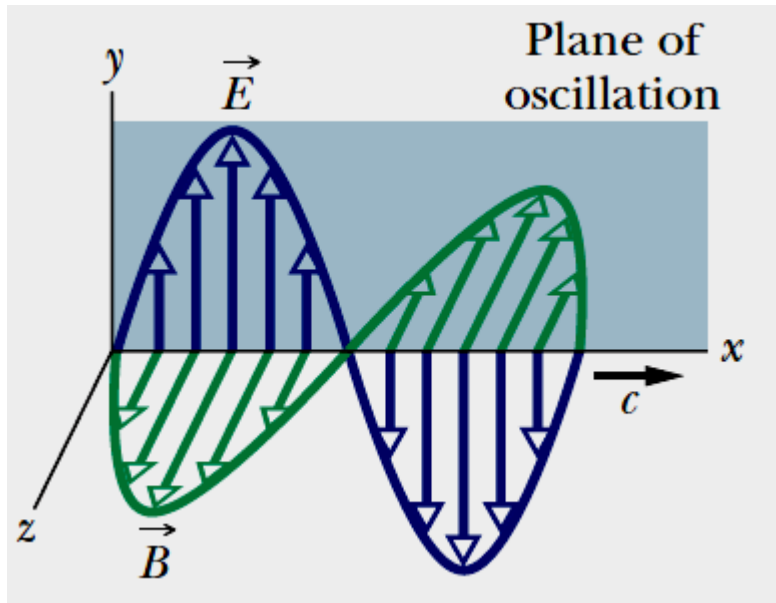
Spaceship powered by radiation pressure



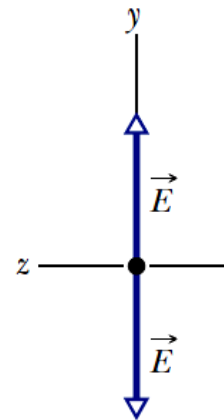
<http://news.sohu.com/20160414/n444128855.shtml>

<http://blog.sciencenet.cn/blog-1319915-970322.html>

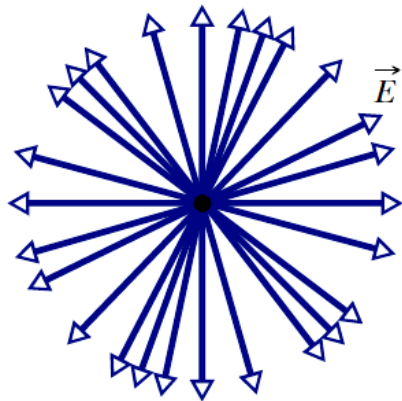
III. Polarization



We normally use the plane of ***E-field*** oscillation as ***the plane of oscillation*** of the wave

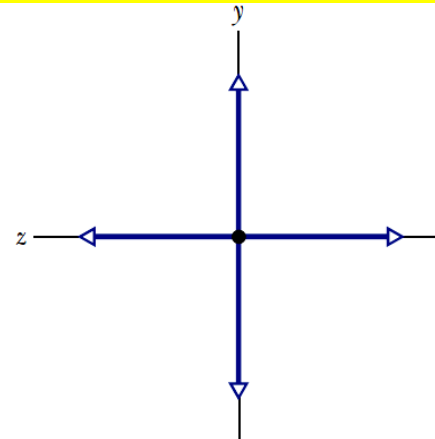


Linearly polarized light

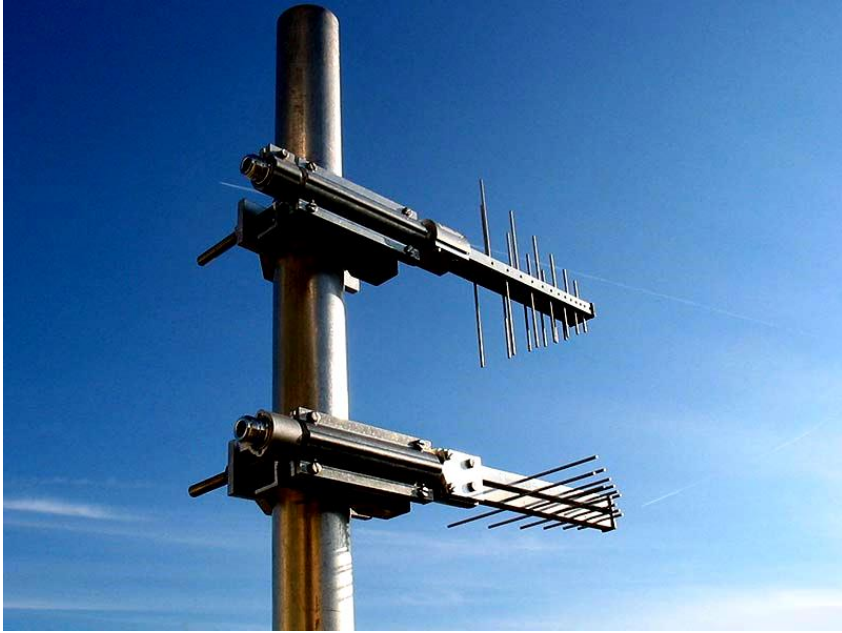


Randomly polarized or unpolarized or natural light

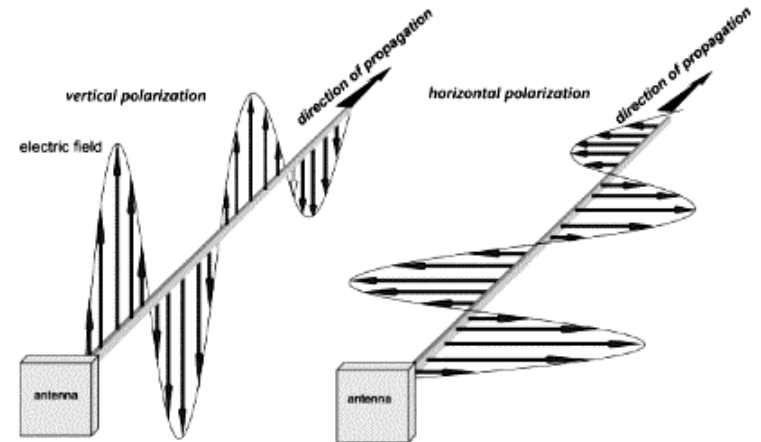
We normally resolved the unpolarized light into 2 components



How to produce polarized EM waves / Light ?



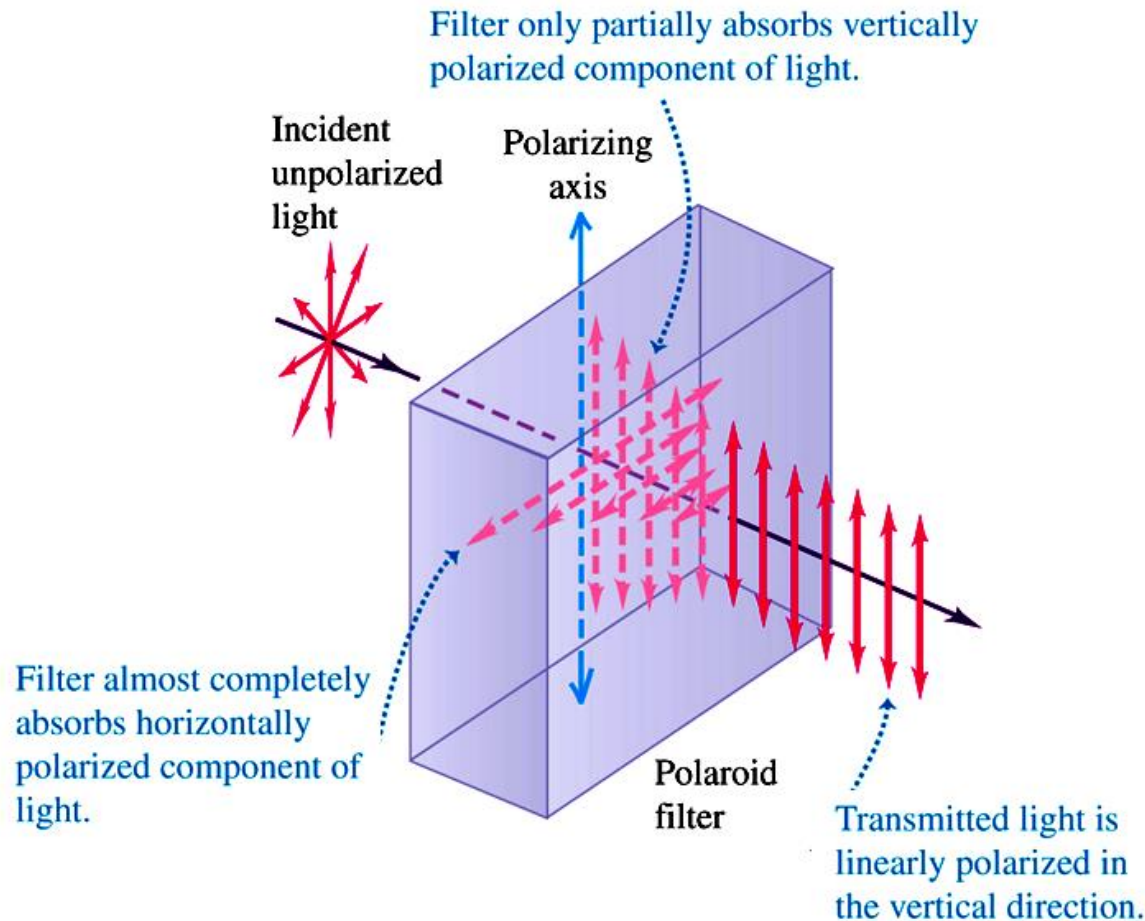
Antenna for polarized sources



Polarized source

How to produce polarized EM waves / Light ?

Unpolarized light loses half of its intensity when passing through a polarizer and becomes linearly polarized.



Polarizing filters / polarizer / Polaroid (for visible light)

Polarizer for EM waves

$$E \rightarrow E_y = E_0 \cos \theta$$

$$I(\propto E^2) \rightarrow I_y = I \cos^2 \theta$$

Unpolarized source

Metal wire

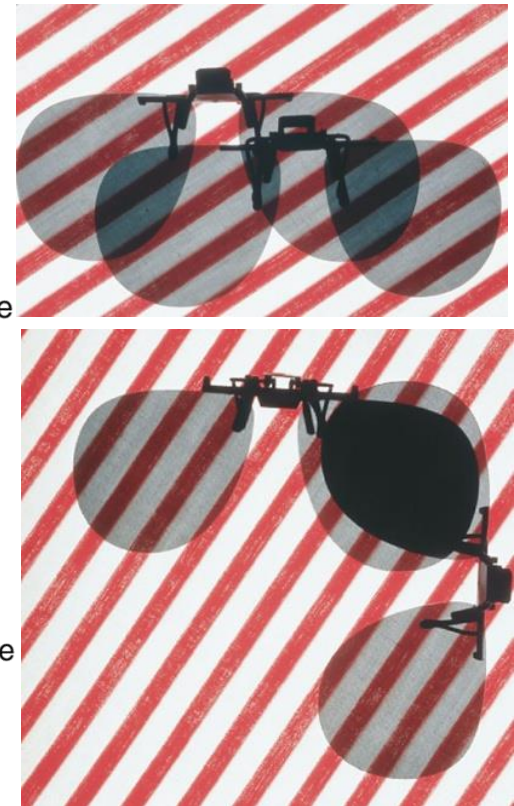
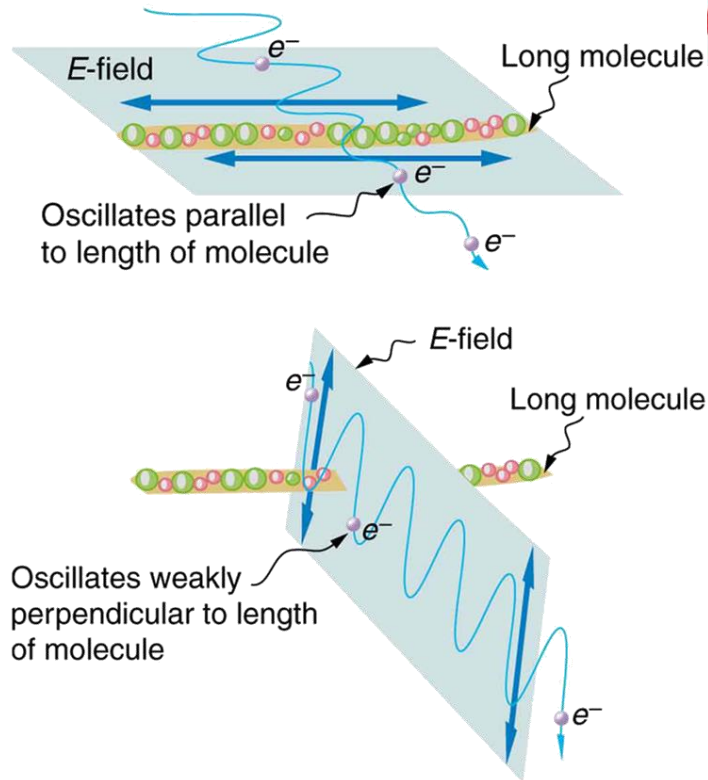
electrons will response to the ***E-field*** which is parallel to the wire and are free to move along the wires

Energy is dissipated by ***I²R*** heating
Perpendicular component is unaffected

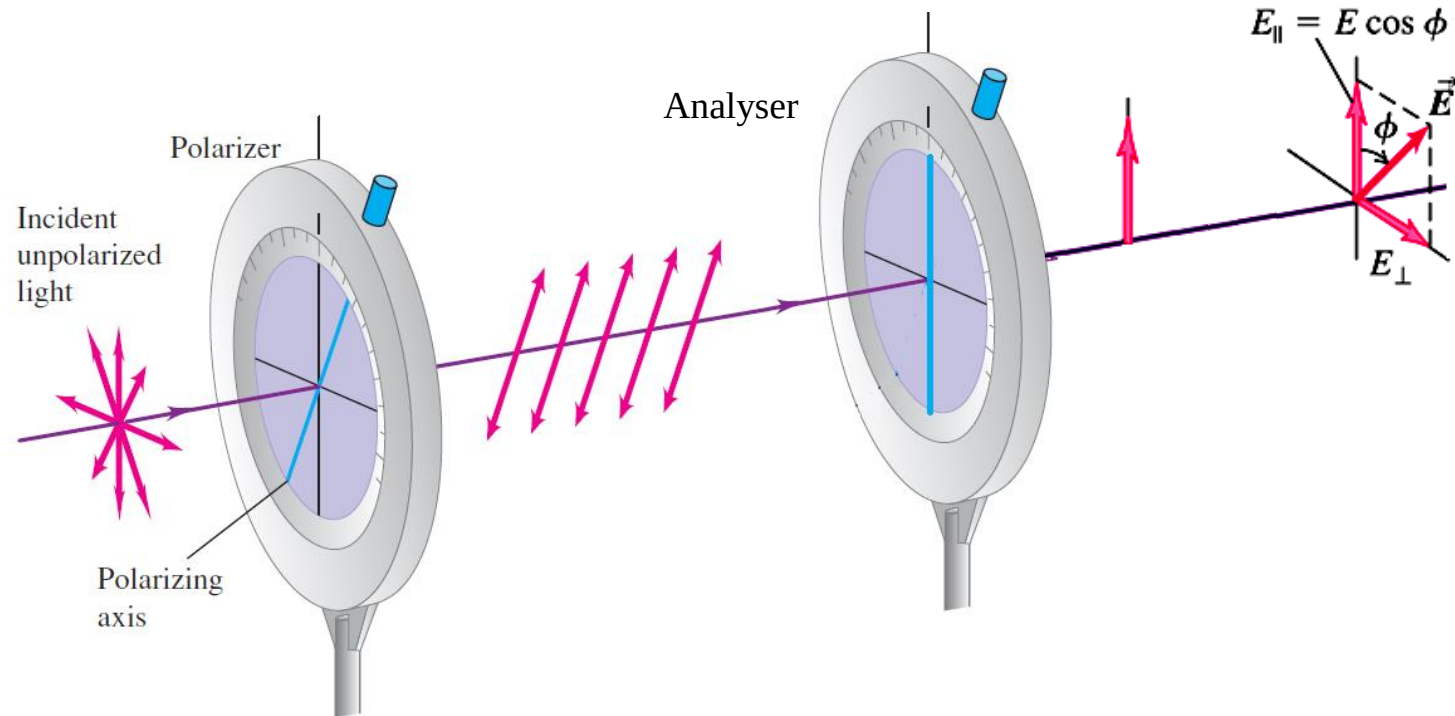
$$\int I_0 \cos^2 \theta d\theta = \frac{1}{2} I_0$$

Polarizing sheets / polaroid for visible light

Edwin Herbert Land (1909 –1991) was an American scientist and inventor. He invented polarized sheets in 1923, and was the co-founder of the Polaroid Corporation.



E-field and Intensity after two polarizers



$$\text{Intensity} = I_0$$

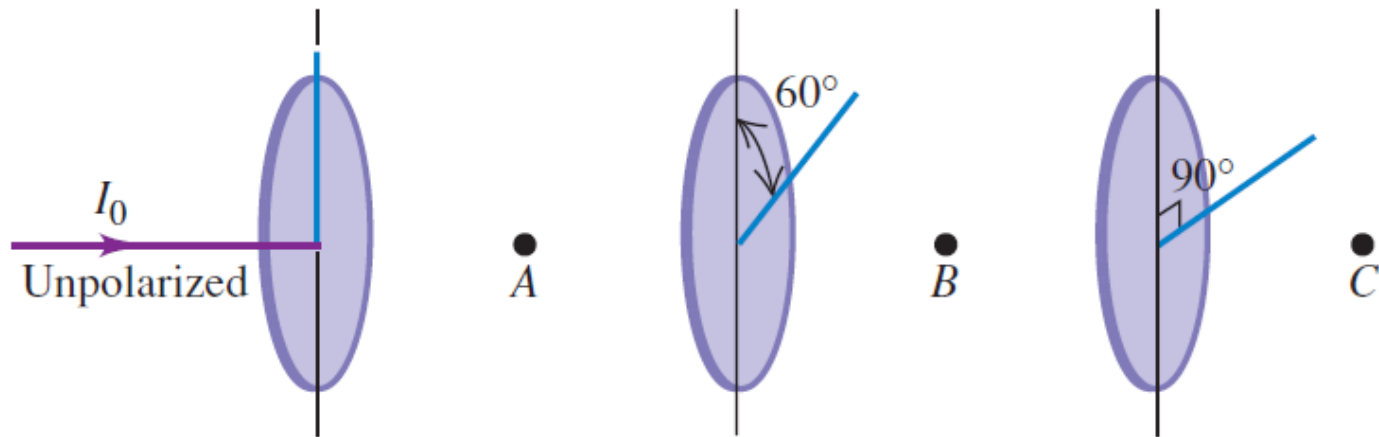
$$\text{Intensity } I_1 = I_0 / 2$$

$$\text{E-field} = E_{\parallel} = E \cos \phi$$

$$\text{E-field} = E$$

$$\text{Intensity } I_2 = I_1 \cos^2 \phi$$

Example



Question:

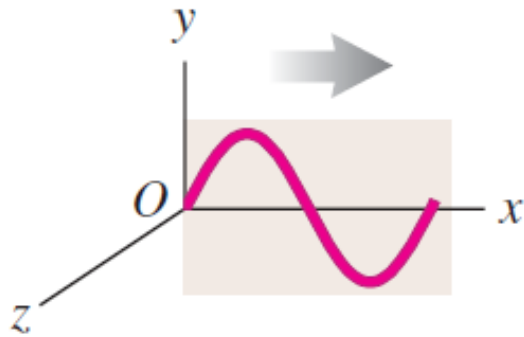
1. Intensity at **A**, **B**, **C**?
2. If middle polarizer is removed, intensity at **C** ?

Answers:

$$A : I_0/2 \quad B : (I_0/2)(\cos 60^\circ)^2 = 0.125I_0 \quad C : (0.125I_0)(\cos 30^\circ)^2 = 0.0938I_0$$

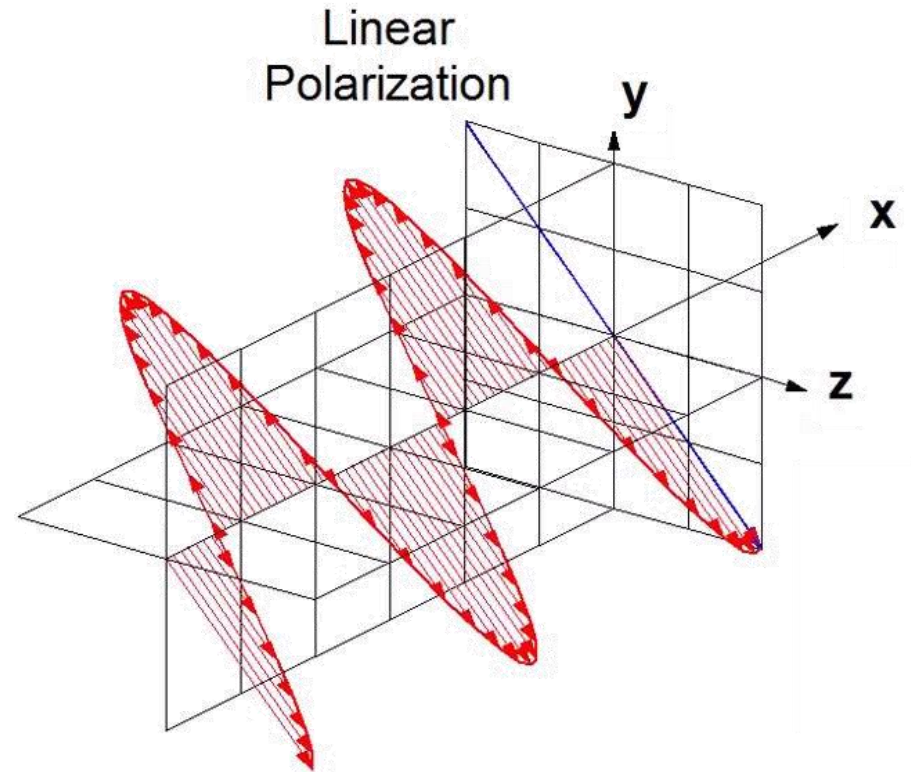
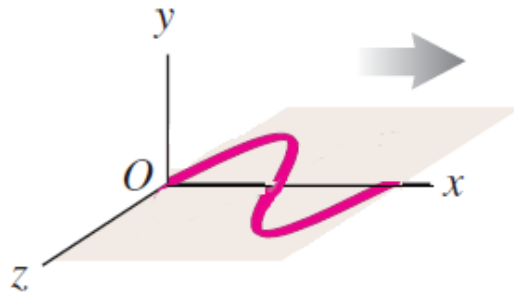
If middle polarizer is removed: $\phi = 90^\circ$ and $I = 0$

Linearly polarized light

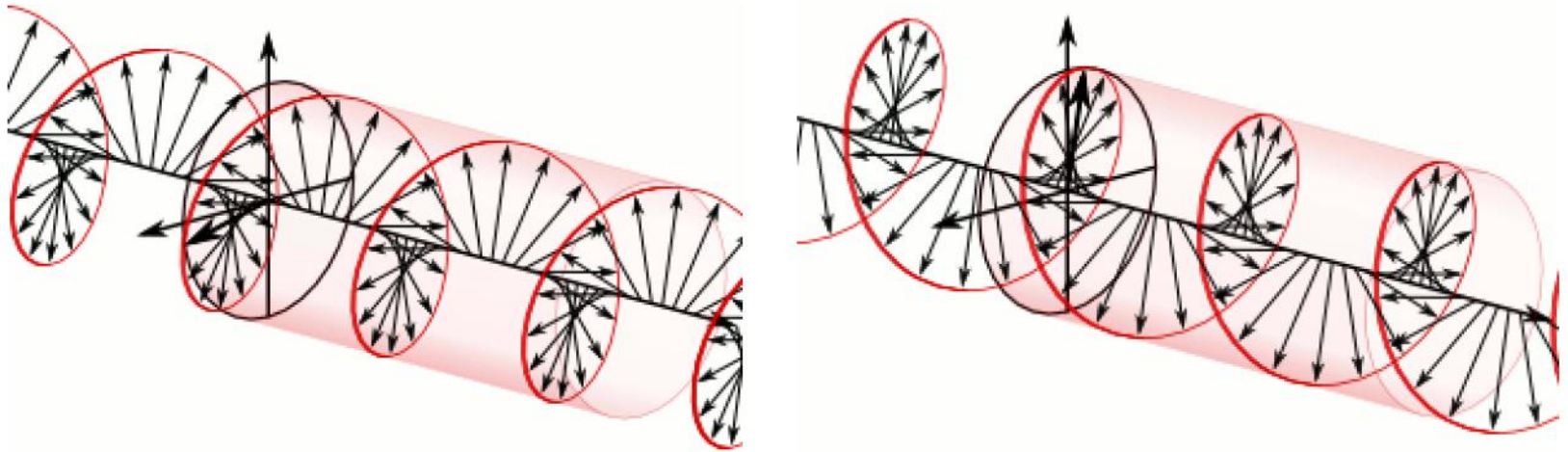


+

=



Circular polarized light



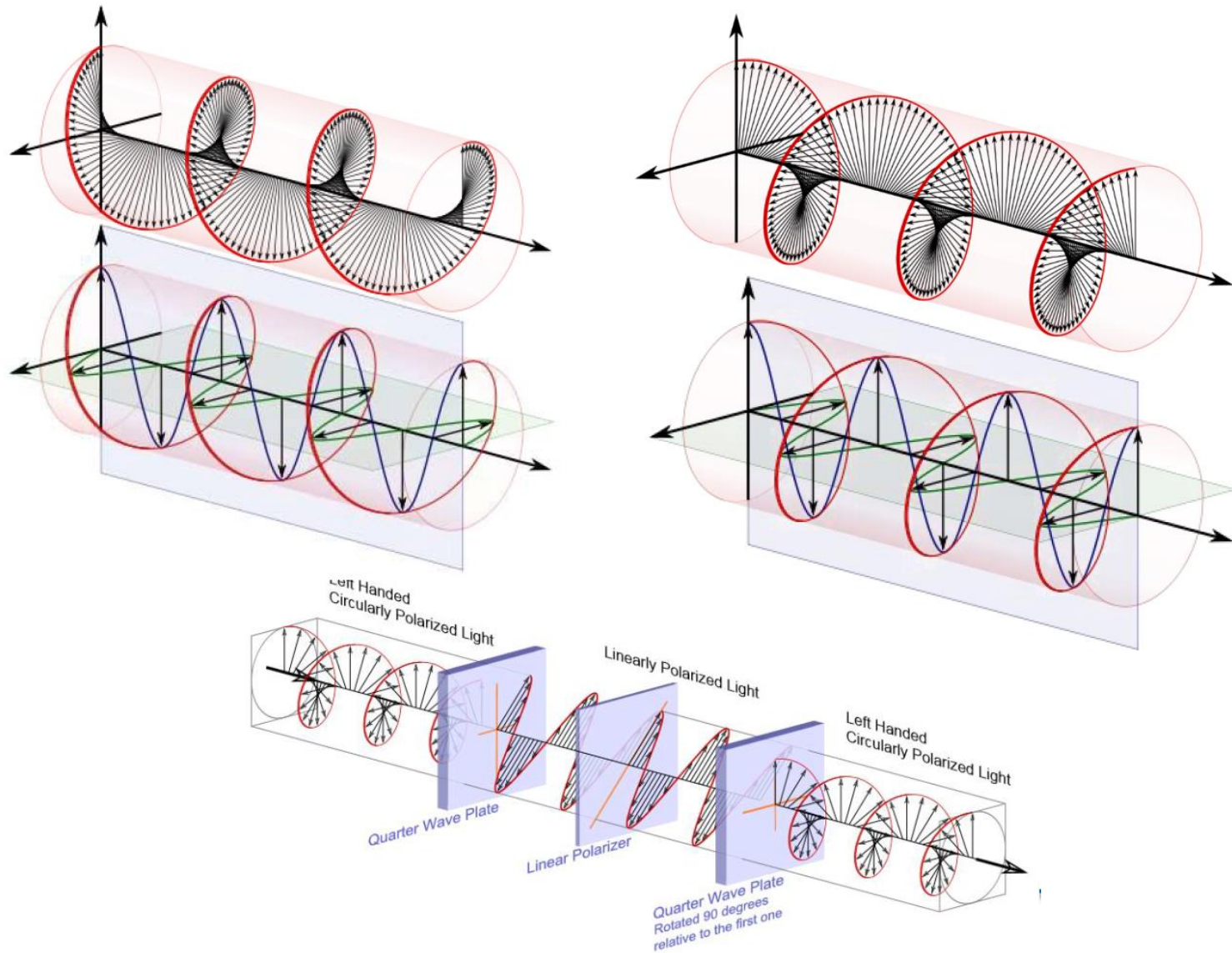
Circular polarization

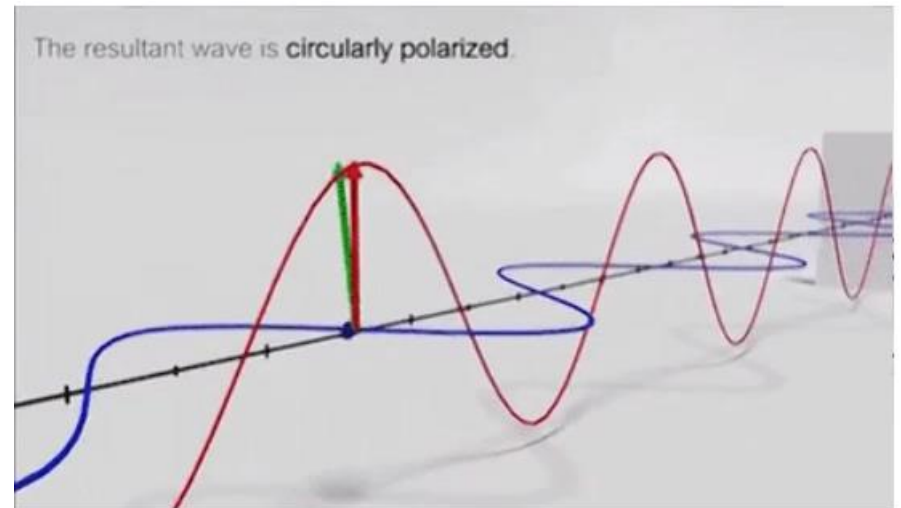
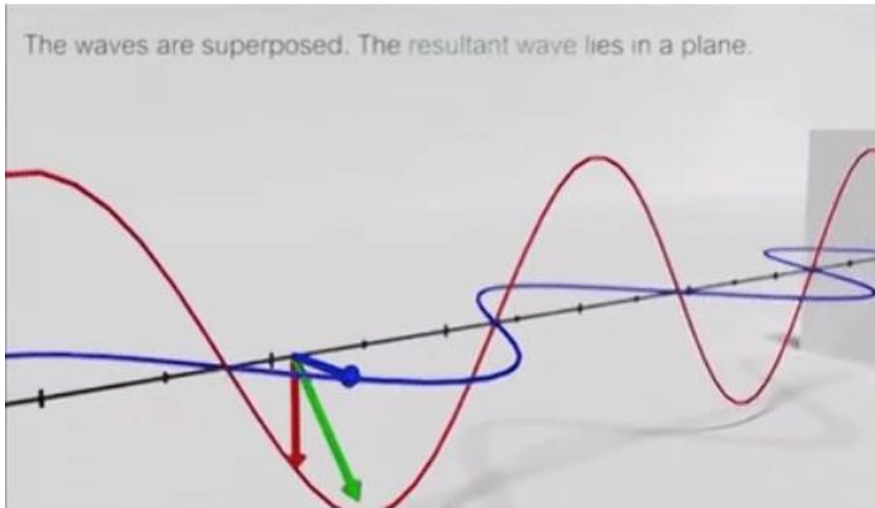
of an electromagnetic wave is a polarization in which the electric field of the passing wave does not change strength but only changes direction in a rotary manner. It can also be understood as a E vector whose x and y components change sinusoidally, and differ in phase by $\pi/2$.

If this wave were approaching an observer, its electric vector appear to be rotating clockwise.

This is called **left-circular** polarization.

Circular polarized light

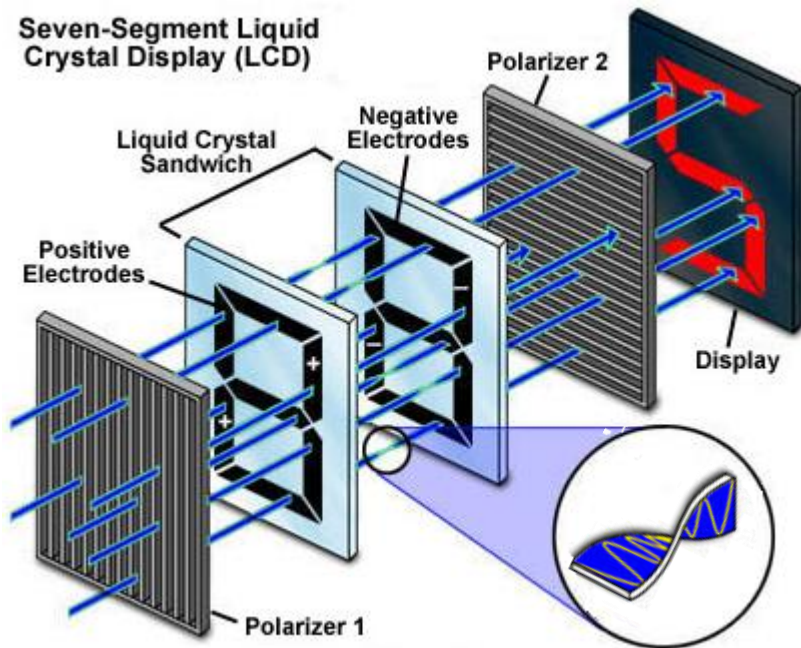




四分之一波片（quarter-wave plate）一定厚度的双折射单晶波片。当光从法向入射透过波片时，[寻常光](#)（o光）和[非常光](#)（e光）之间的位相差等于 $\pi/2$ 或其奇数倍，这样的晶片称为四分之一波片或 1/4波片。当线偏振光垂直入射1/4波片,并且[光的偏振](#)和波片的光轴面（垂直自然裂开面）成 θ 角，出射后成[椭圆偏振光](#)。特别当 $\theta=45^\circ$ 时，出射光为圆偏振光。波片的快轴和慢轴，与晶体的类型有关。负晶体的 $V_e > V_o$ ，波片光轴方向平行于波片平面，负晶体做的四分之一波片的光轴方向就是快轴方向。正晶体快轴方向垂直于光轴方向位于玻片平面内。

为什么1/4波片可以用于隔离反射光呢？激光器发出的经过偏振片选择后的线偏振光，经过1/4波片后，两个分量产生1/4波长相位差，反射回来后，两个分量的方向和相位差不变，再次经过1/4波片，两个分量再次叠加1/4波长相位差，共1/2相位差，所以合成的光为线偏振光，且偏振方向与原方向垂直，所以无法再次通过偏振片

Polarization in Seven-Segment LCD



Light passing through polarizer 1 is polarized in the vertical direction and, when no current is applied to the electrodes, the liquid crystalline phase induces a 90 degree "twist" of the light and it can pass through polarizer 2

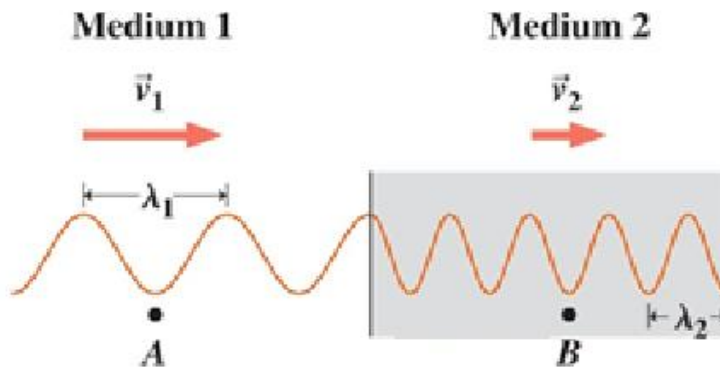
When current is applied to the electrodes, the liquid crystalline phase aligns with the current and light passing through a charged electrode is not twisted and is blocked by polarizer 2.

Index of refraction (折射率)

Wave speeds differ in different media.

$$n = \frac{c}{v}$$

index of refraction



$$\frac{E_m}{B_m} = c \quad (\text{amplitude ratio}) \quad \frac{qvB}{qE} = \frac{v}{c}$$

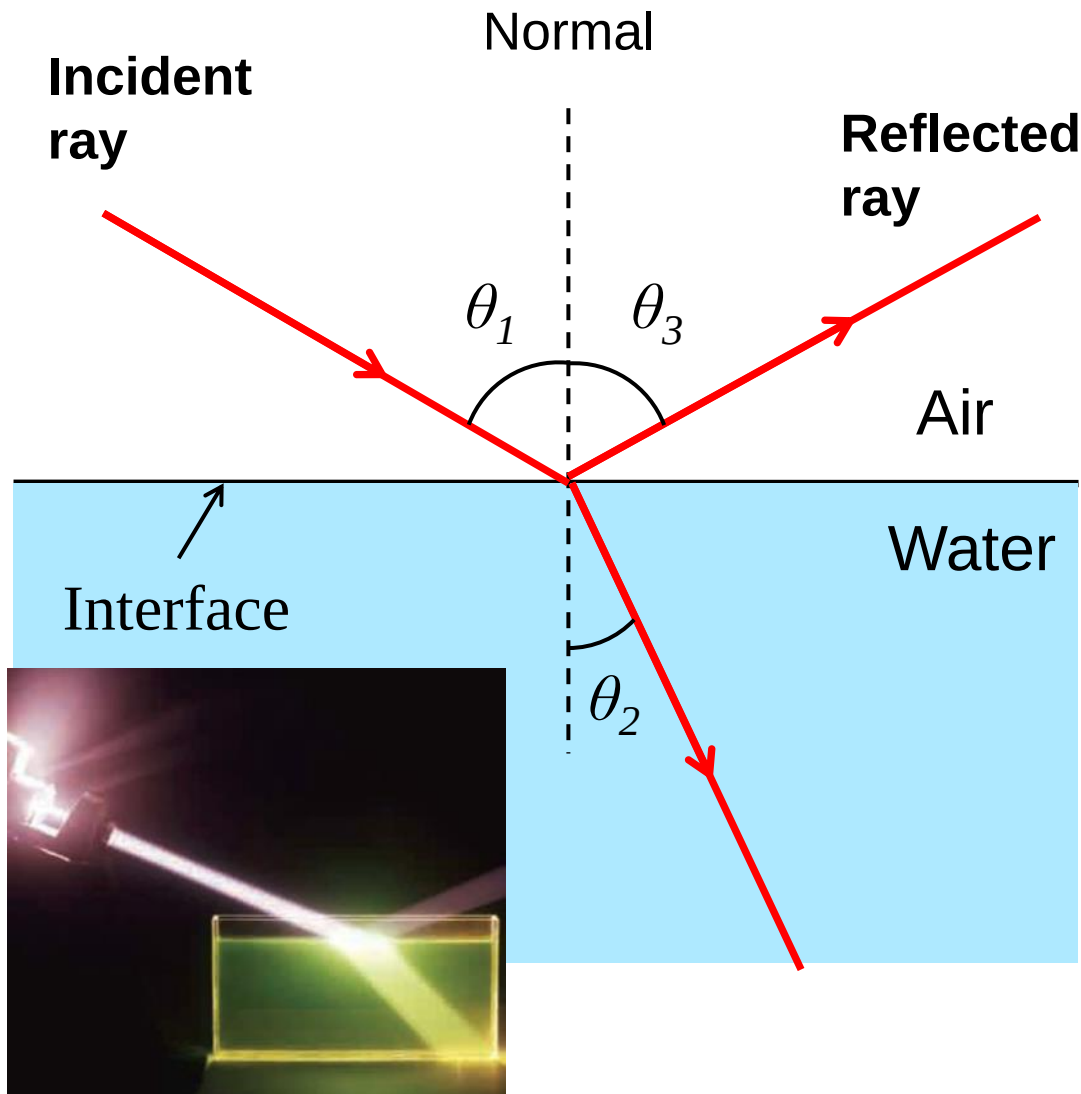
Observers at A & B count same number of wave crests at any given duration

→ wave frequency doesn't change when crossing media.

$$\lambda = \frac{v}{f} = \frac{c}{n f}$$

→ λ smaller for medium with slower v or higher n .

IV. Reflection (反射) and Refraction (折射)



(1) The three rays are on the same plane

$$(2) \theta_1 = \theta_3$$

Law of refraction (**Snell's Law**)

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_{\text{water}}}{n_{\text{air}}} = \frac{n_2}{n_1} = 1.3$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Index of refraction n $n = c / v$

$n \sim 1.0003$ air

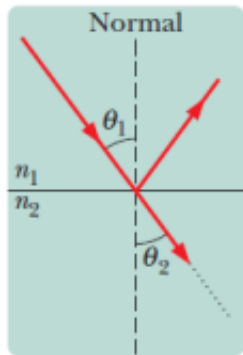
~ 1.333 water

$\sim 1.5 - 2$ glass

$$n = \frac{\lambda_{\text{vacuum}}}{\lambda_{\text{medium}}}$$

Reflection and refraction

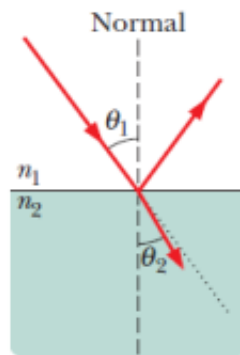
$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$



$n_2 = n_1$

(a)

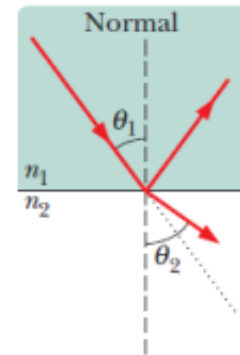
If the indexes match, there is no direction change.



$n_2 > n_1$

(b)

If the next index is greater, the ray is bent *toward* the normal.



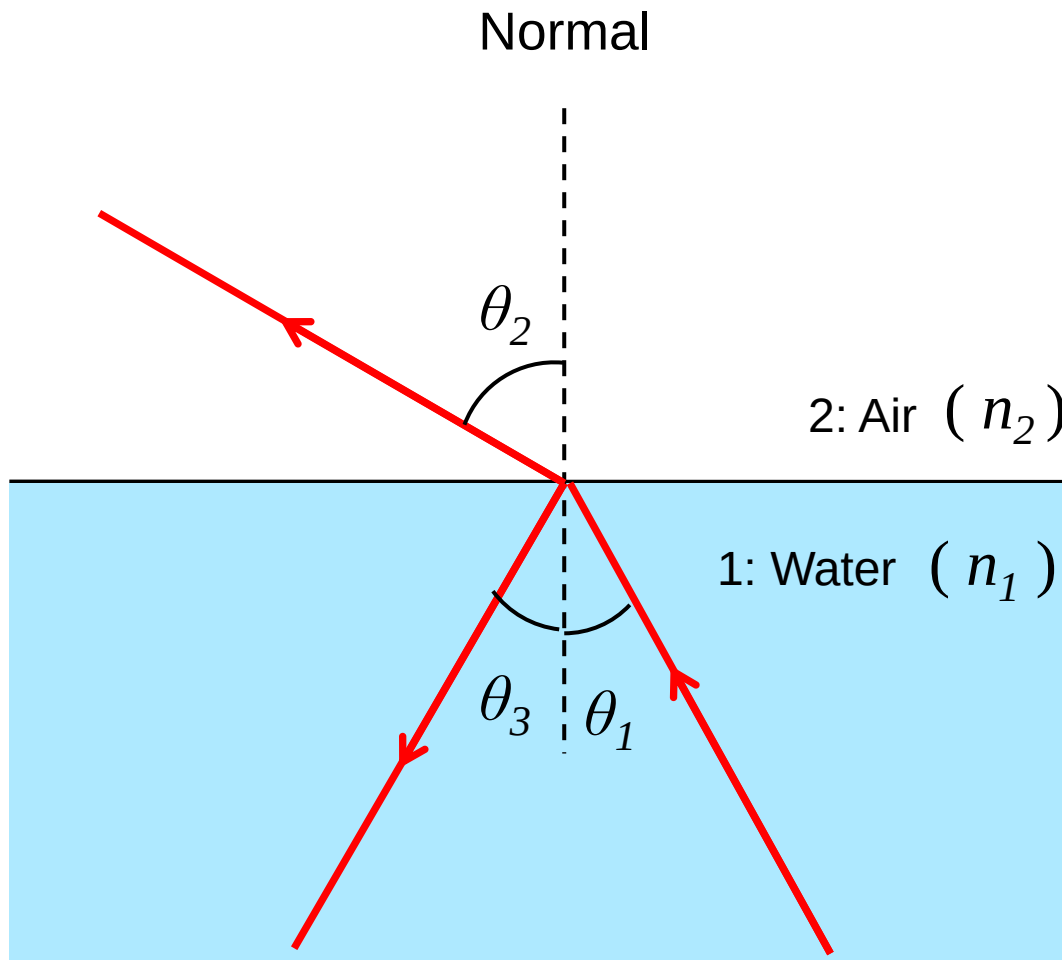
$n_2 < n_1$

(c)

If the next index is less, the ray is bent *away from* the normal.

$$n_i = \frac{c}{v_i}$$

Reflection (反射) and Refraction (折射)



$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_{\text{air}}}{n_{\text{water}}} = \frac{1}{1.3}$$

$$\text{If } \theta_2 = 90^\circ \rightarrow \theta_1 = 50^\circ$$

This is called **critical angle**

$$\text{If } \theta_1 > 50^\circ$$

Total internal reflection

$$\sin \theta_{\text{critical}} = \frac{n_2}{n_1}$$

Only when $n_2 < n_1$

Total internal refraction

If light goes from a high- n material to a low- n one, and the incident angle is large enough, no light will be refracted out.

According to Snell's law,

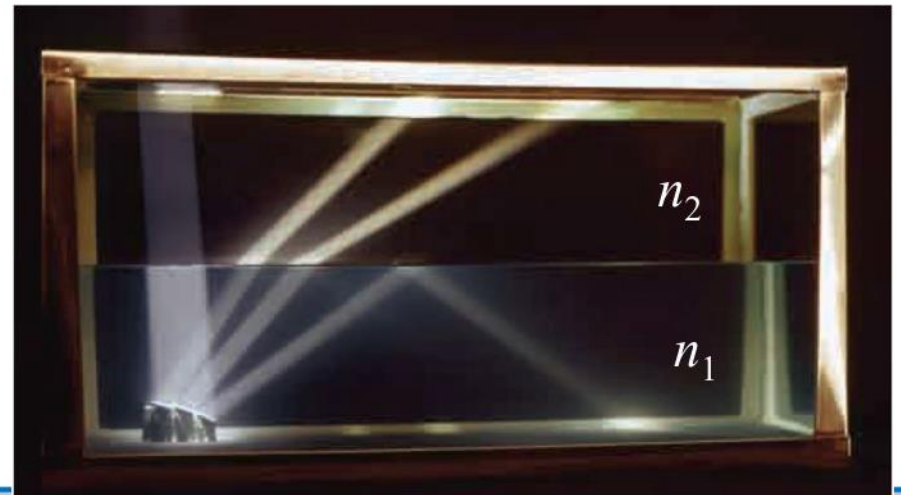
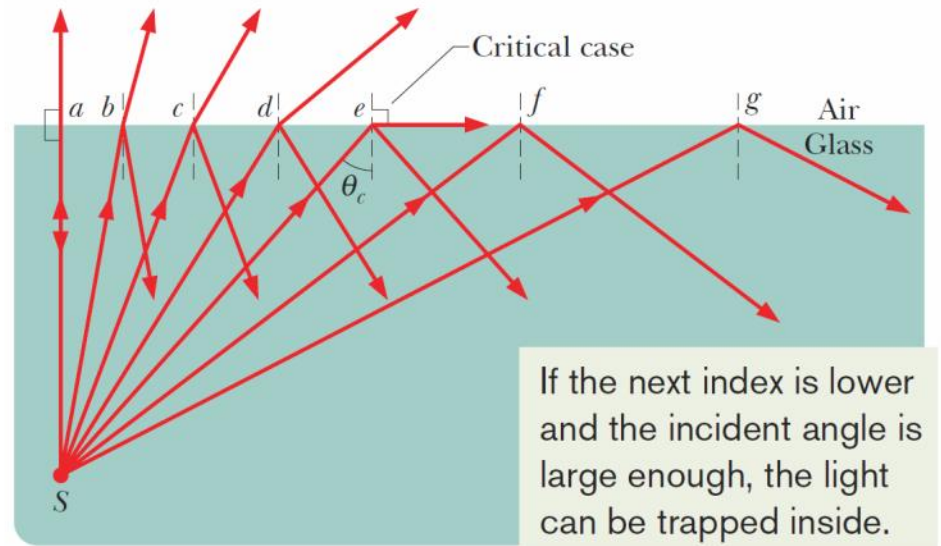
$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$

So the critical condition for total internal refraction is

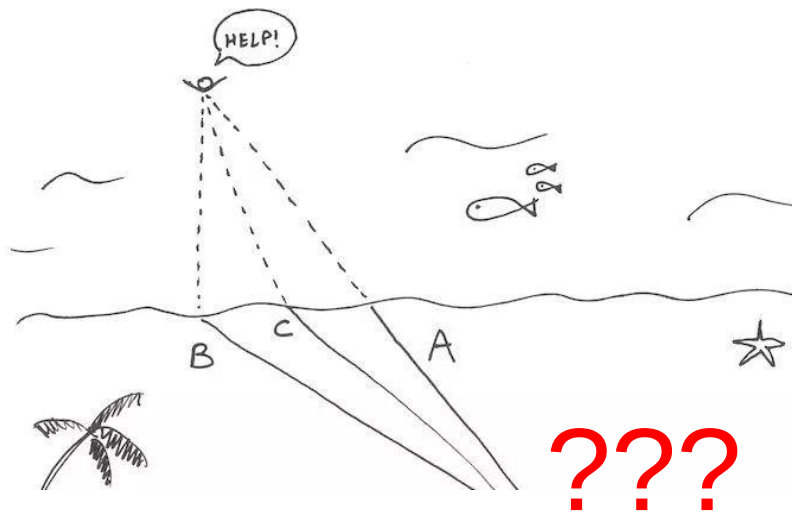
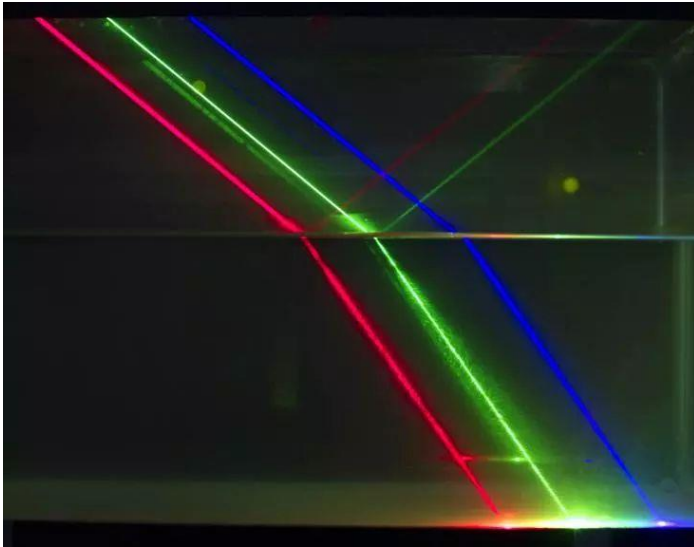
$$\frac{n_1}{n_2} \sin \theta_c = 1,$$

or

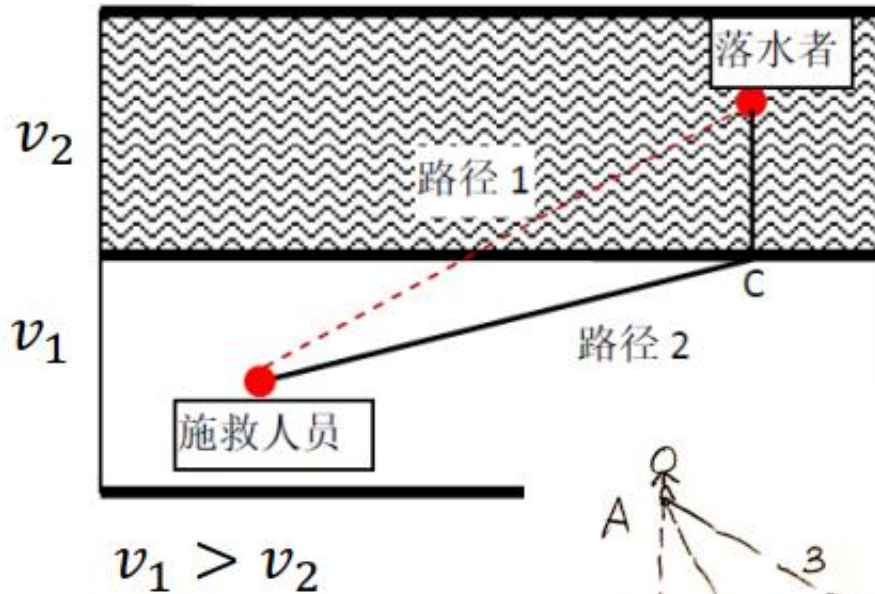
$$\theta_c = \sin^{-1} \frac{n_2}{n_1}.$$



General Physics II, SUSTC

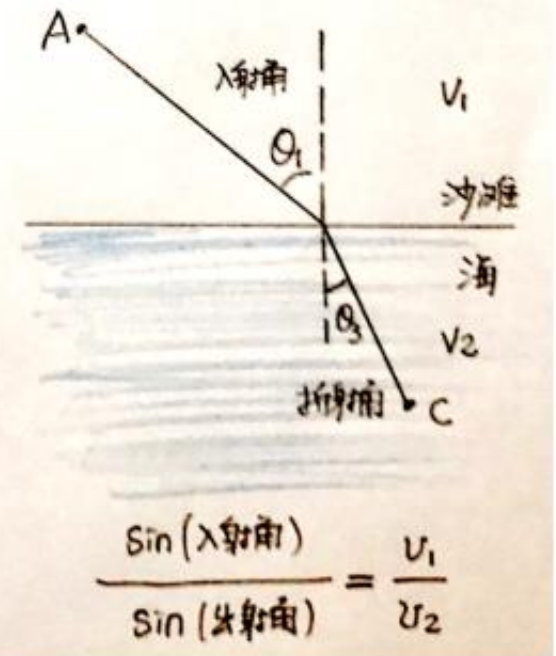
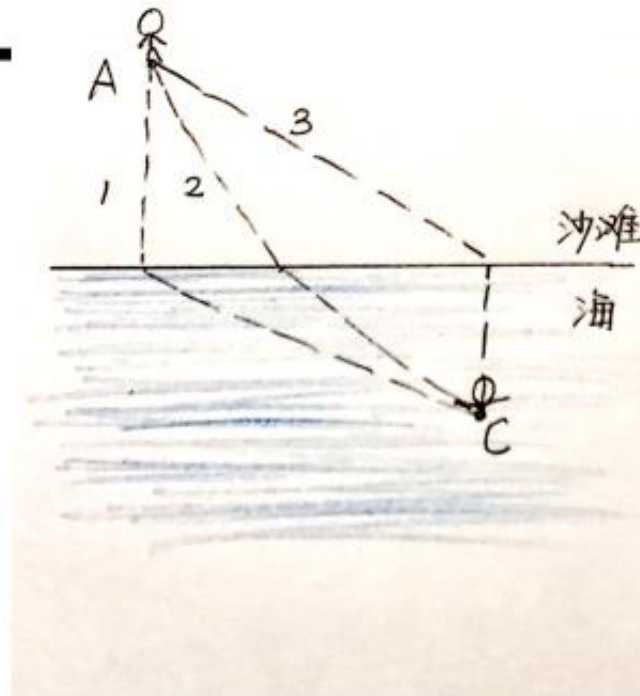


Optimization problem



Fastest way?

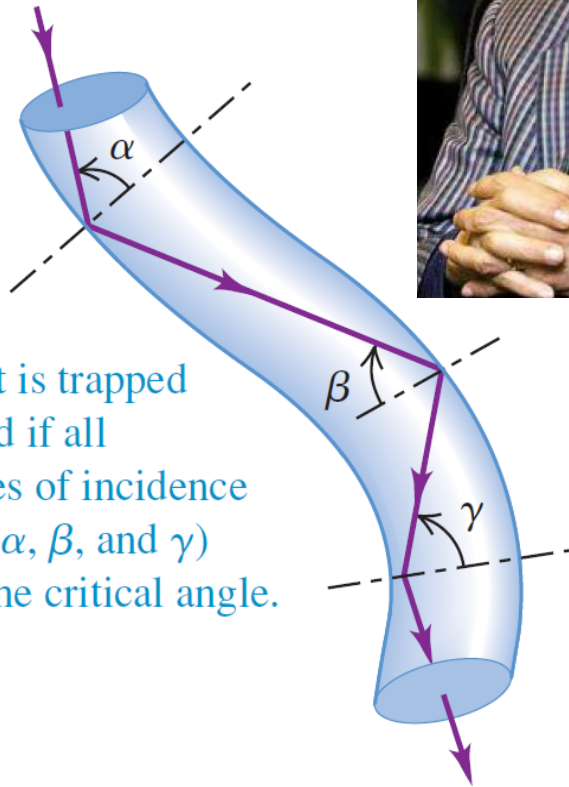
Law of refraction !



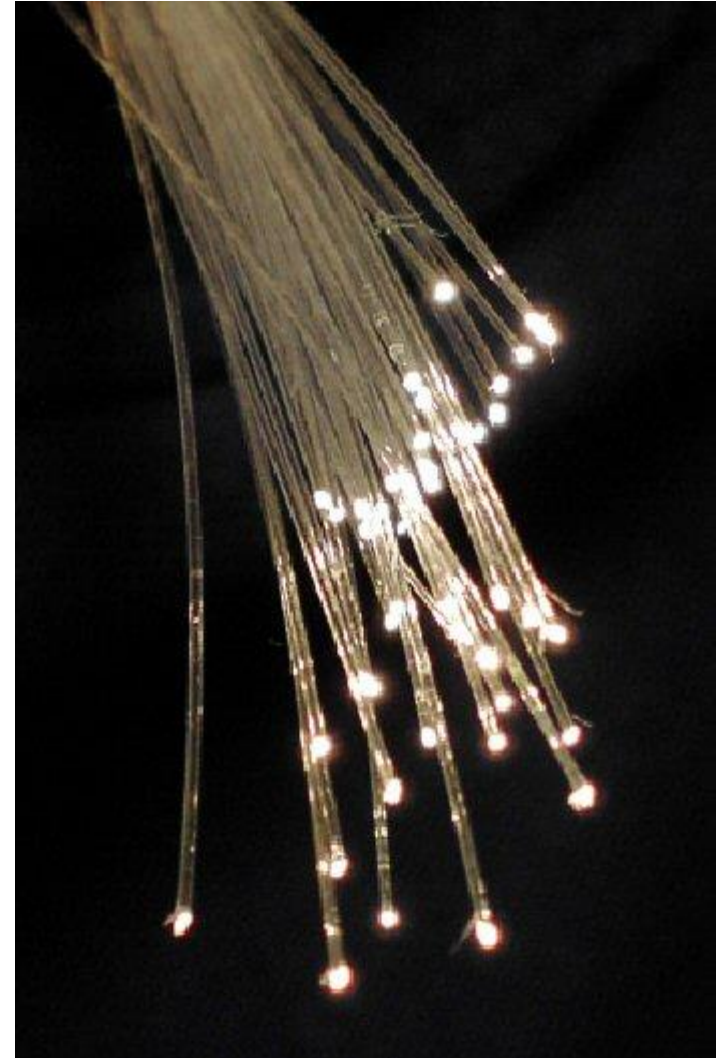
$$\frac{\sin(\text{入射角})}{\sin(\text{折射角})} = \frac{v_1}{v_2}$$

Application: Optical Fiber

Charles Kuen Gao
(1933- 2018)
Nobel Prize 2009



The light is trapped in the rod if all the angles of incidence (such as α , β , and γ) exceed the critical angle.



Index of refraction (折射率)

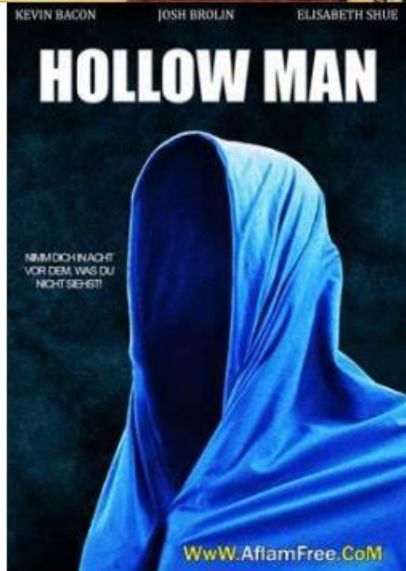
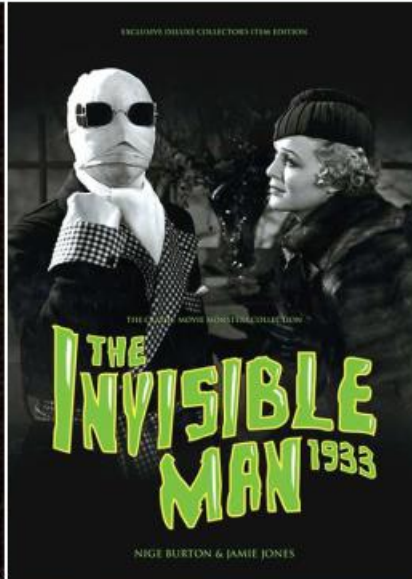
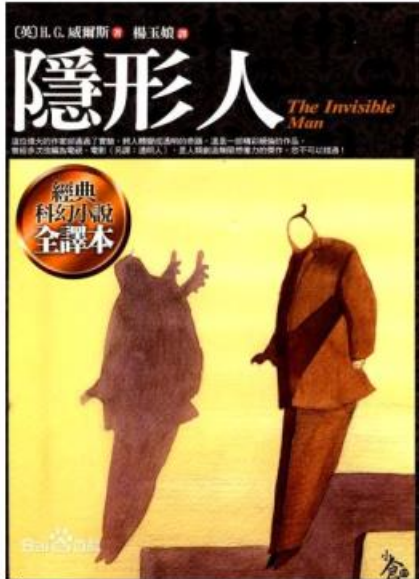
Some Indexes of Refraction^a

Medium	Index	Medium	Index
Vacuum	Exactly 1	Typical crown glass	1.52
Air (STP) ^b	1.00029	Sodium chloride	1.54
Water (20°C)	1.33	Polystyrene	1.55
Acetone	1.36	Carbon disulfide	1.63
Ethyl alcohol	1.36	Heavy flint glass	1.65
Sugar solution (30%)	1.38	Sapphire	1.77
Fused quartz	1.46	Heaviest flint glass	1.89
Sugar solution (80%)	1.49	Diamond	2.42

^aFor a wavelength of 589 nm (yellow sodium light).

^bSTP means “standard temperature (0°C) and pressure (1 atm).”

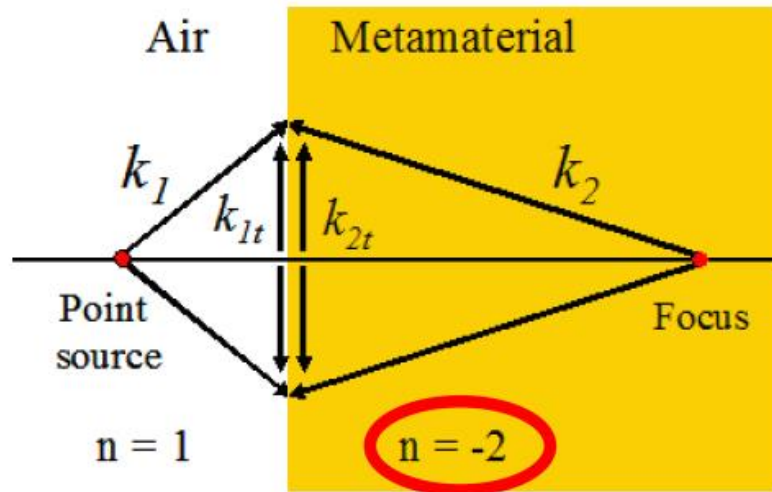
How to make an Invisible Person (隱形人)



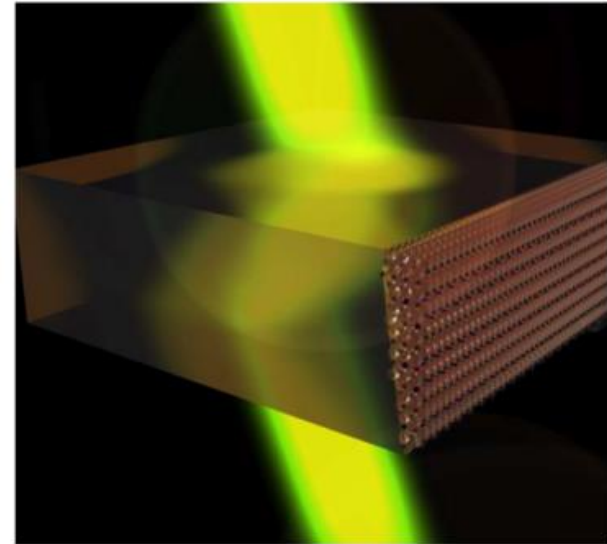
<https://baike.baidu.com/item/%E9%9A%90%E5%BD%A2%E4%BA%BA/4860>

<https://haokan.baidu.com/v?vid=15000488306232841707&pd=bjh&fr=bjhaauthor&type=video>

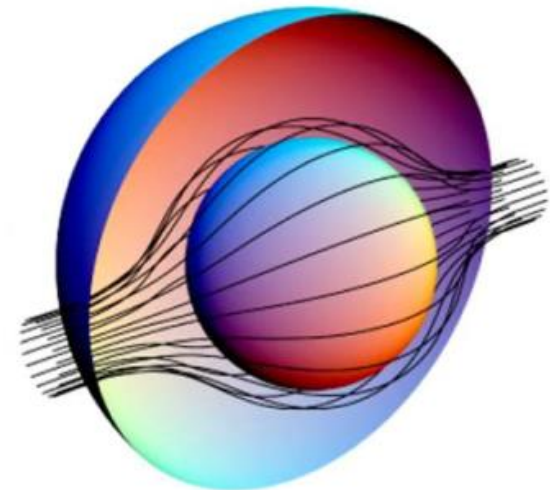
Can the refractive index be less than zero?



Reflection and refraction

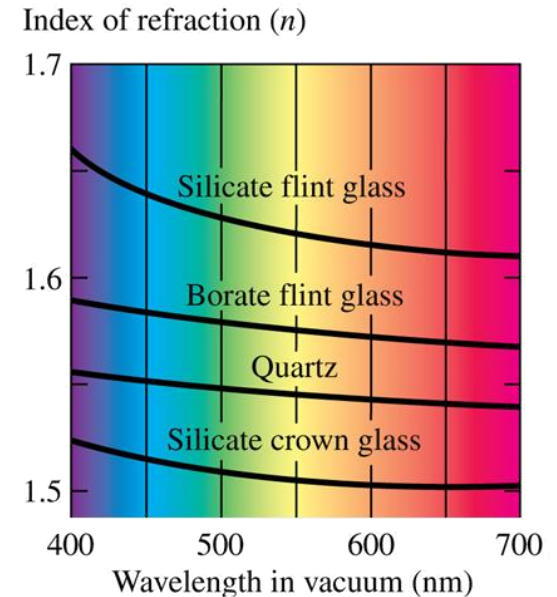
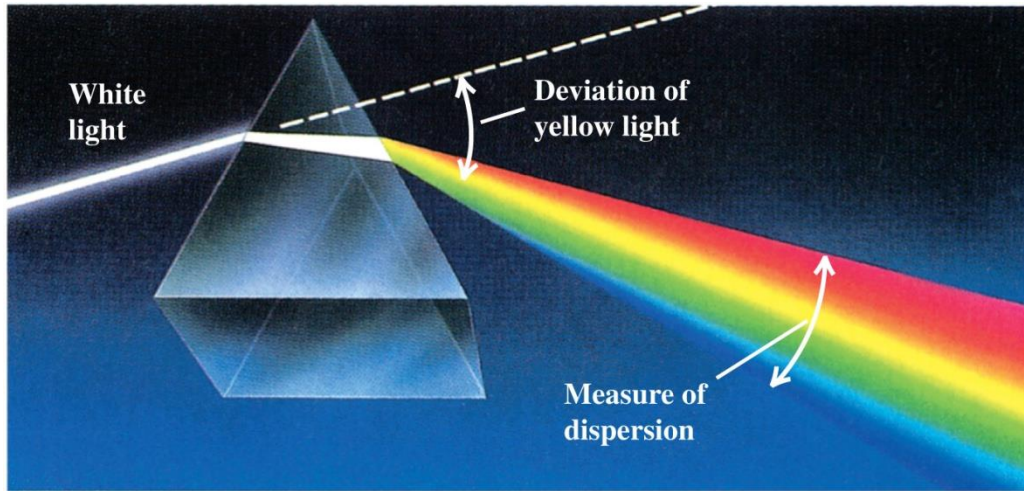


Invisible Cloak



<https://haokan.baidu.com/v?vid=15000488306232841707&pd=bjh&fr=bjhaauthor&type=video>

V. Chromatic Dispersion 色散



Index of refraction **n** depends on wavelength!

$$n = \frac{c}{v}$$

Speed of light in vacuum is the same for all wavelengths

Speed of light in a material is different for different wavelengths

Rainbows → refraction + reflection + dispersion



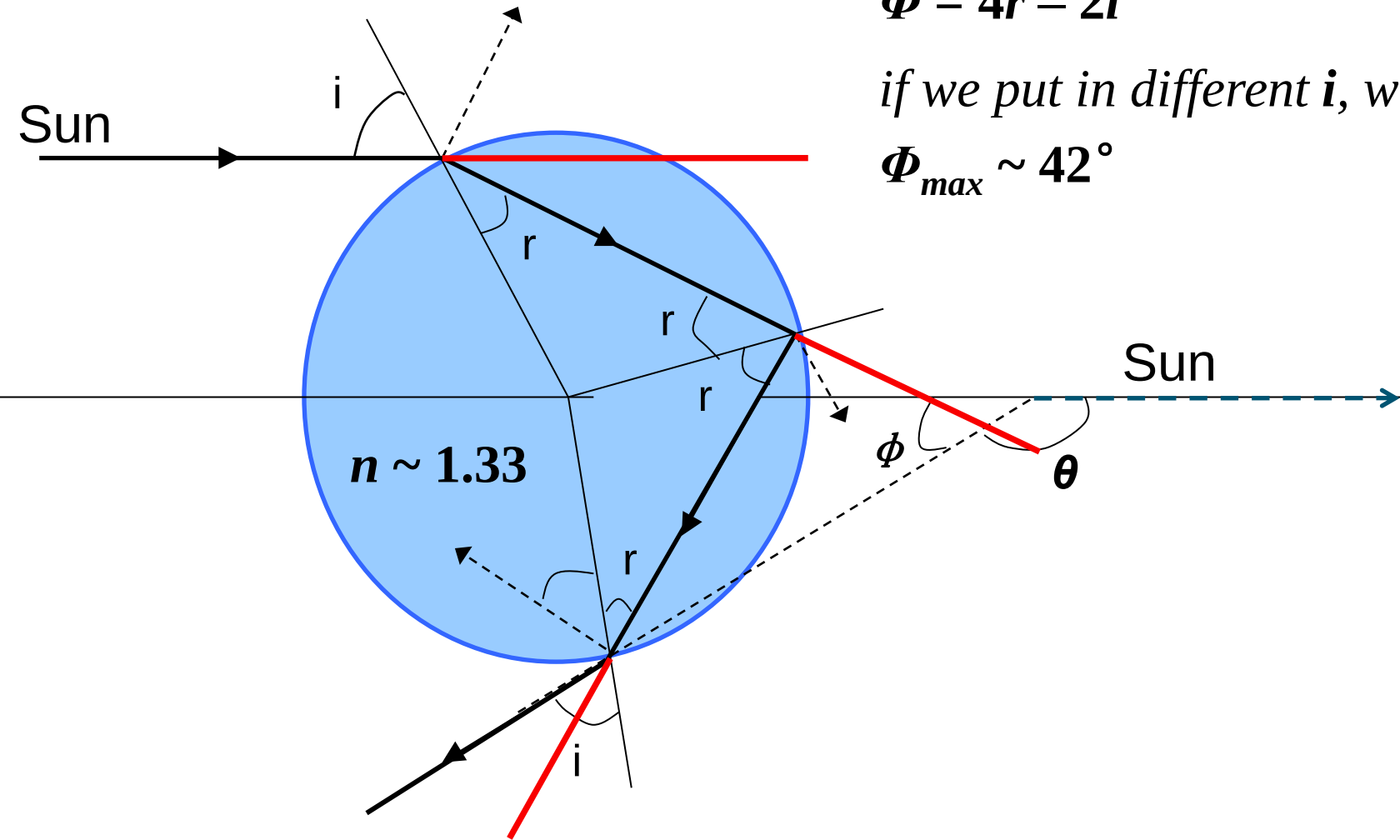
Light in a raindrop

$$\theta = 180 + 2i - 4r$$

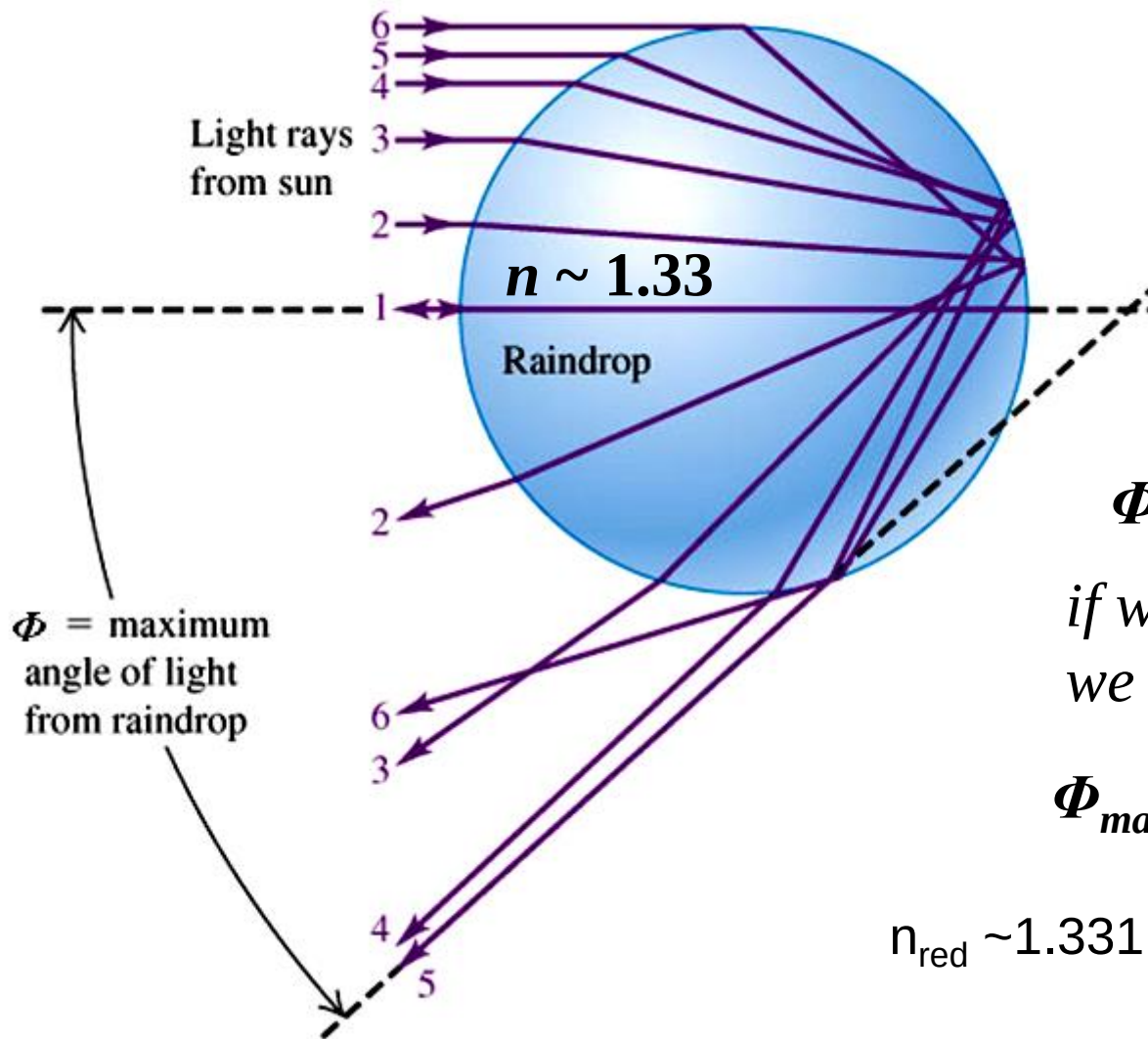
$$\Phi = 4r - 2i$$

if we put in different i , we found

$$\Phi_{\max} \sim 42^\circ$$



Light in a raindrop



$$\Phi = 4r - 2i$$

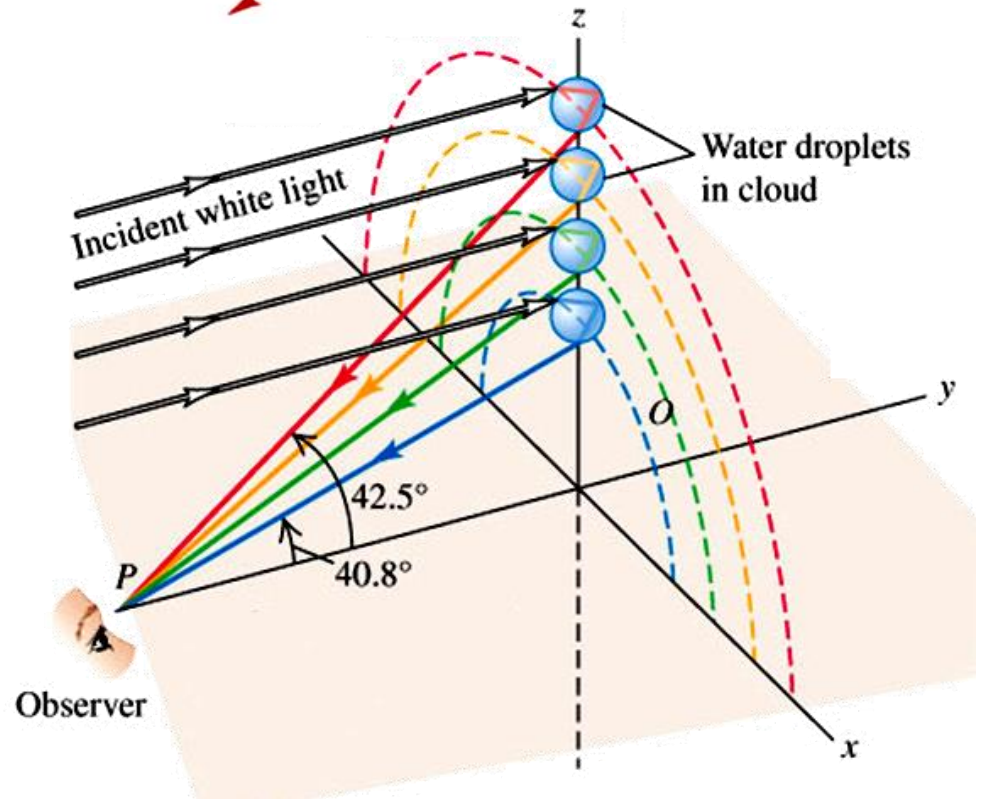
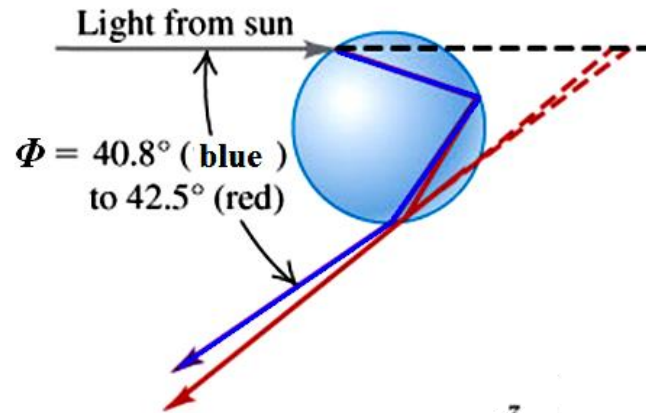
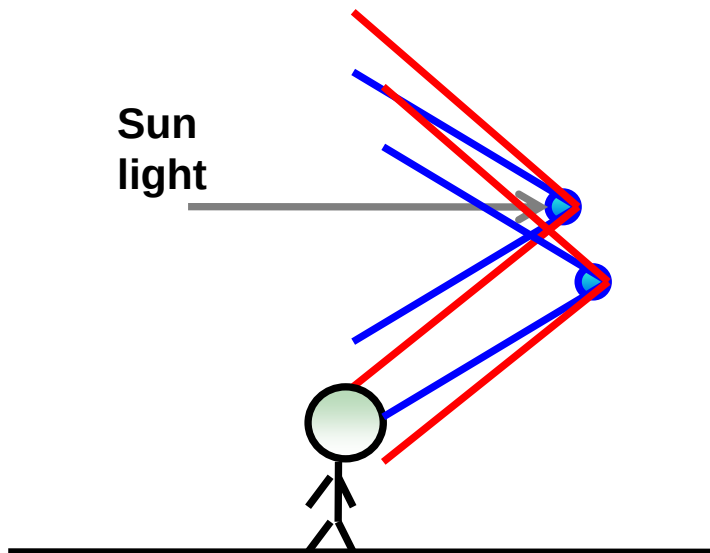
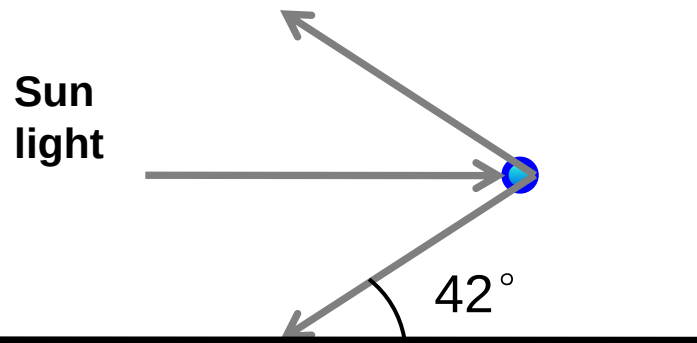
if we put in different i ,
we found

$$\Phi_{\max} \sim 42^\circ$$

$$n_{\text{red}} \sim 1.331 \quad \Phi_{\max} \sim 42.5^\circ$$

$$n_{\text{blue}} \sim 1.343 \quad \Phi_{\max} \sim 40.8^\circ$$

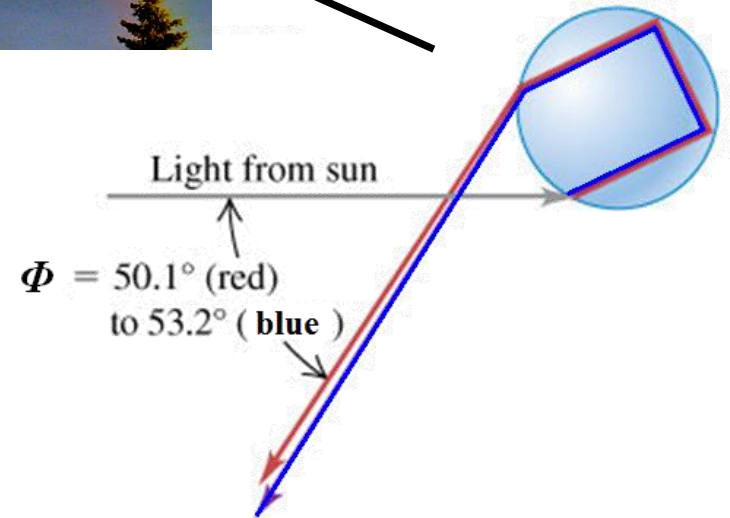
Light in a raindrop



Primary and secondary rainbow



If the sky is clear enough, you can see a fainter, secondary rainbow (霓) outside the primary one (虹). Light goes through a different refractive path to form the secondary rainbow.



When will you see a rainbow?

Rainbow at the Grand Canyon



And rainbows everywhere



Rainbow



Rainbow at Las Vegas, photo by Jinlong Zhu



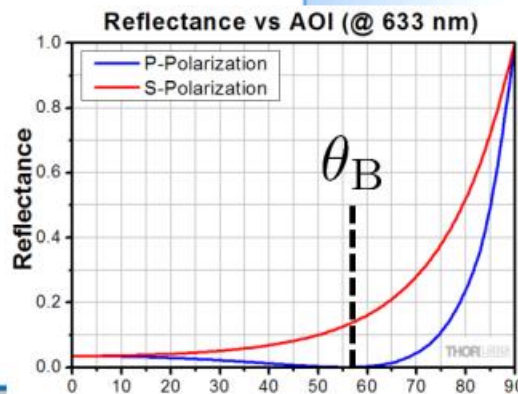
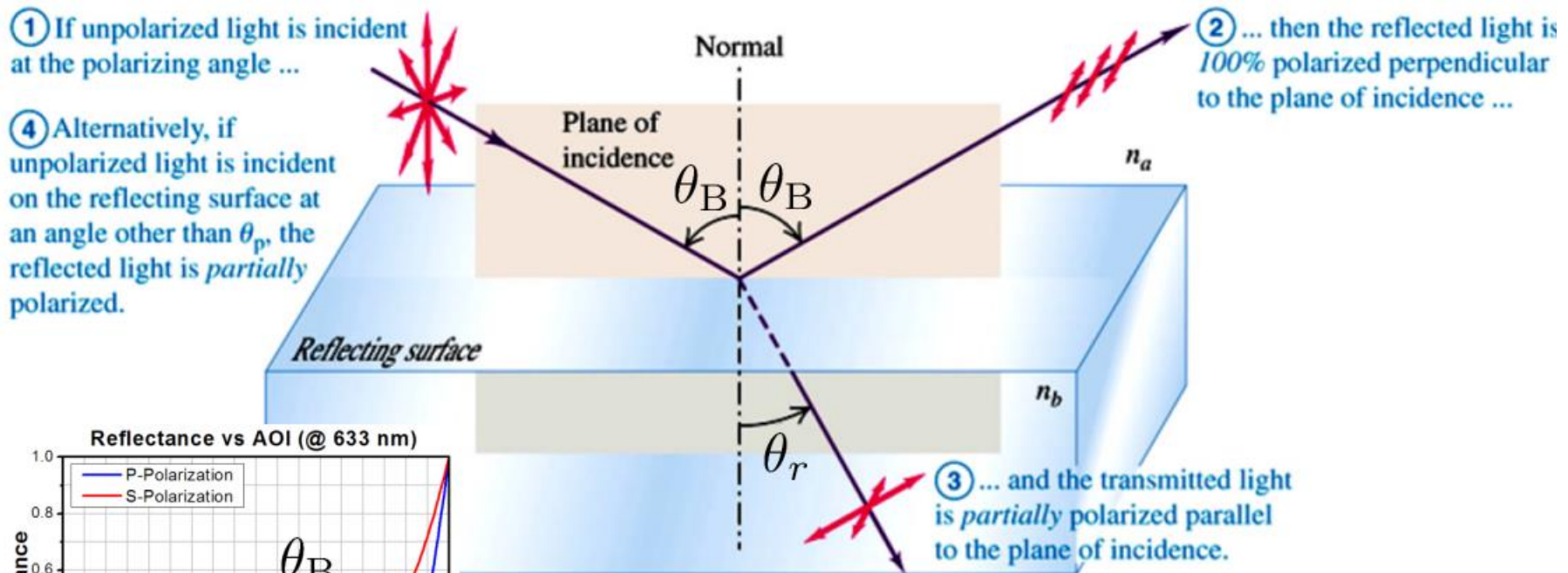
VI. Polarization by reflection

In 1812, Sir David Brewster found by experiment that at certain incident angle, we called it polarization angle or **Brewster angle**, the following things happen:

① If unpolarized light is incident at the polarizing angle ...

④ Alternatively, if unpolarized light is incident on the reflecting surface at an angle other than θ_p , the reflected light is *partially* polarized.

② ... then the reflected light is 100% polarized perpendicular to the plane of incidence ...



P-polarization : E-field parallel to the plane of incidence

S-polarization : E-field perpendicular to the plane of incidence

Polarization by reflection

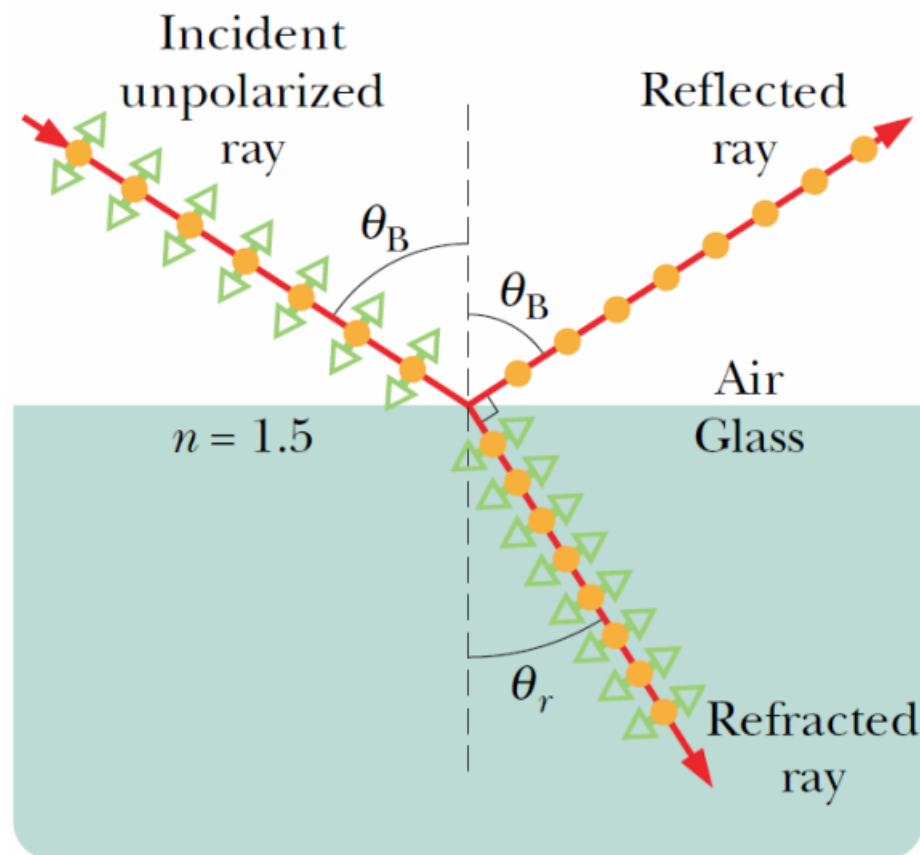
It is found experimentally that, for light incident at the Brewster angle, the *reflected light* is **perpendicular** to the *refracted light*,

$$\theta_B + \theta_r = \pi/2.$$

According to Snell's law, for light coming from air into a media with refractive index n , the Brewster angle is

$$\theta_B = \tan^{-1} \frac{n_2}{n_1}$$

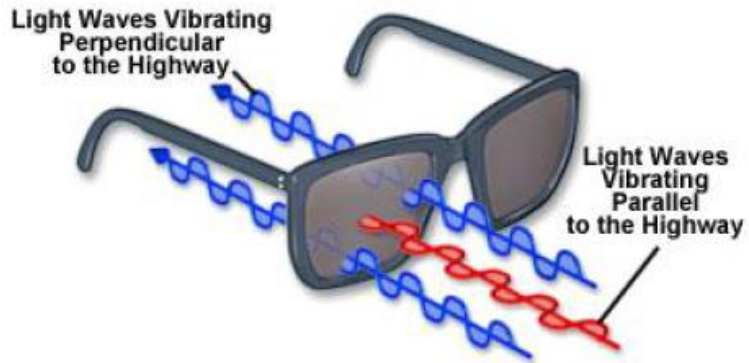
(Derive yourself!)



- Component perpendicular to page
- ↔ Component parallel to page

Polarization by reflection

Polarized glasses for driving



Thru a Standard Lens

Thru a Polarized Lens



WITHOUT POLARIZED



WITH POLARIZED



WITHOUT POLARIZED



WITH POLARIZED



Electromagnetic Waves An electromagnetic wave consists of oscillating electric and magnetic fields. The various possible frequencies of electromagnetic waves form a *spectrum*, a small part of which is visible light. An electromagnetic wave traveling along an x axis has an electric field $\vec{E}$ and a magnetic field $\vec{B}$ with magnitudes that depend on x and t :

$$E = E_m \sin(kx - \omega t)$$

$$\text{and} \quad B = B_m \sin(kx - \omega t), \quad (33-1, 33-2)$$

where E_m and B_m are the amplitudes of $\vec{E}$ and $\vec{B}$. The oscillating electric field induces the magnetic field, and the oscillating magnetic field induces the electric field. The speed of any electromagnetic wave in vacuum is c , which can be written as

$$c = \frac{E}{B} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}, \quad (33-5, 33-3)$$

where E and B are the simultaneous (but nonzero) magnitudes of the two fields.

Energy Flow The rate per unit area at which energy is transported via an electromagnetic wave is given by the Poynting vector $\vec{S}$:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}. \quad (33-19)$$

The direction of $\vec{S}$ (and thus of the wave's travel and the energy transport) is perpendicular to the directions of both $\vec{E}$ and $\vec{B}$. The time-averaged rate per unit area at which energy is transported is S_{avg} , which is called the *intensity* I of the wave:

$$I = \frac{1}{c\mu_0} E_{\text{rms}}^2, \quad (33-26)$$

in which $E_{\text{rms}} = E_m/\sqrt{2}$. A *point source* of electromagnetic waves emits the waves *isotropically*—that is, with equal intensity in all directions. The intensity of the waves at distance r from a point source of power P_s is

$$I = \frac{P_s}{4\pi r^2}. \quad (33-27)$$

Radiation Pressure When a surface intercepts electromagnetic radiation, a force and a pressure are exerted on the surface. If the radiation is totally absorbed by the surface, the force is

$$F = \frac{IA}{c} \quad (\text{total absorption}), \quad (33-32)$$

in which I is the intensity of the radiation and A is the area of the surface perpendicular to the path of the radiation. If the radiation is totally reflected back along its original path, the force is

$$F = \frac{2IA}{c} \quad (\text{total reflection back along path}). \quad (33-33)$$

The radiation pressure p_r is the force per unit area:

$$p_r = \frac{I}{c} \quad (\text{total absorption}) \quad (33-34)$$

and

$$p_r = \frac{2I}{c} \quad (\text{total reflection back along path}). \quad (33-35)$$

Polarization Electromagnetic waves are **polarized** if their electric field vectors are all in a single plane, called the *plane of oscillation*. From a head-on view, the field vectors oscillate parallel to a single axis perpendicular to the path taken by the waves. Light waves from common sources are not polarized; that is, they are **unpolarized**, or **polarized randomly**. From a head-on view, the vectors oscillate parallel to every possible axis that is perpendicular to the path taken by the waves.

Polarizing Sheets When a polarizing sheet is placed in the path of light, only electric field components of the light parallel to the sheet's **polarizing direction** are *transmitted* by the sheet; components perpendicular to the polarizing direction are absorbed. The light that emerges from a polarizing sheet is polarized parallel to the polarizing direction of the sheet.

If the original light is initially unpolarized, the transmitted intensity I is half the original intensity I_0 :

$$I = \frac{1}{2} I_0. \quad (33-36)$$

If the original light is initially polarized, the transmitted intensity depends on the angle θ between the polarization direction of the original light (the axis along which the fields oscillate) and the polarizing direction of the sheet:

$$I = I_0 \cos^2 \theta. \quad (33-38)$$

Geometrical Optics *Geometrical optics* is an approximate treatment of light in which light waves are represented as straight-line rays.

Reflection and Refraction When a light ray encounters a boundary between two transparent media, a **reflected** ray and a **refracted** ray generally appear. Both rays remain in the plane of incidence. The **angle of reflection** is equal to the angle of incidence, and the **angle of refraction** is related to the angle of incidence by Snell's law,

$$n_2 \sin \theta_2 = n_1 \sin \theta_1 \quad (\text{refraction}), \quad (33-40)$$

where n_1 and n_2 are the indexes of refraction of the media in which the incident and refracted rays travel.

Total Internal Reflection A wave encountering a boundary across which the index of refraction decreases will experience **total internal reflection** if the angle of incidence exceeds a **critical angle** θ_c , where

$$\theta_c = \sin^{-1} \frac{n_2}{n_1} \quad (\text{critical angle}). \quad (33-45)$$

Polarization by Reflection A reflected wave will be fully **polarized**, with its $\vec{E}$ vectors perpendicular to the plane of incidence, if the incident, unpolarized wave strikes a boundary at the **Brewster angle** θ_B , where

$$\theta_B = \tan^{-1} \frac{n_2}{n_1} \quad (\text{Brewster angle}). \quad (33-49)$$